

Vertebral Column and Spinal Cord

Department Anatomy
Neuroanatomy (4)

First Part

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Lectuer Note:

➤ Introduction and Review of The vertebral column.

- The vertebral column.
- Vertebrae
- Gneral featuers of vertebrae.
- Curvatuers of vertiberal colum.
- Cervical Vertebrae.
- First and Second Cervical Vertebrae.
- The Thoracic Vertebrae.
- The lumbar vertebrae.
- The Joints between vertebrae.
- Ligaments between vertebrae.
- The disc.
- Blood supply of vertebra and spinal cord.
- Nerve Supply of Vertebral Joints.
- Dermatomes innervated by posterior rami of spinal nerves
- Intervertebral foramen

➤ The Spinal Cord

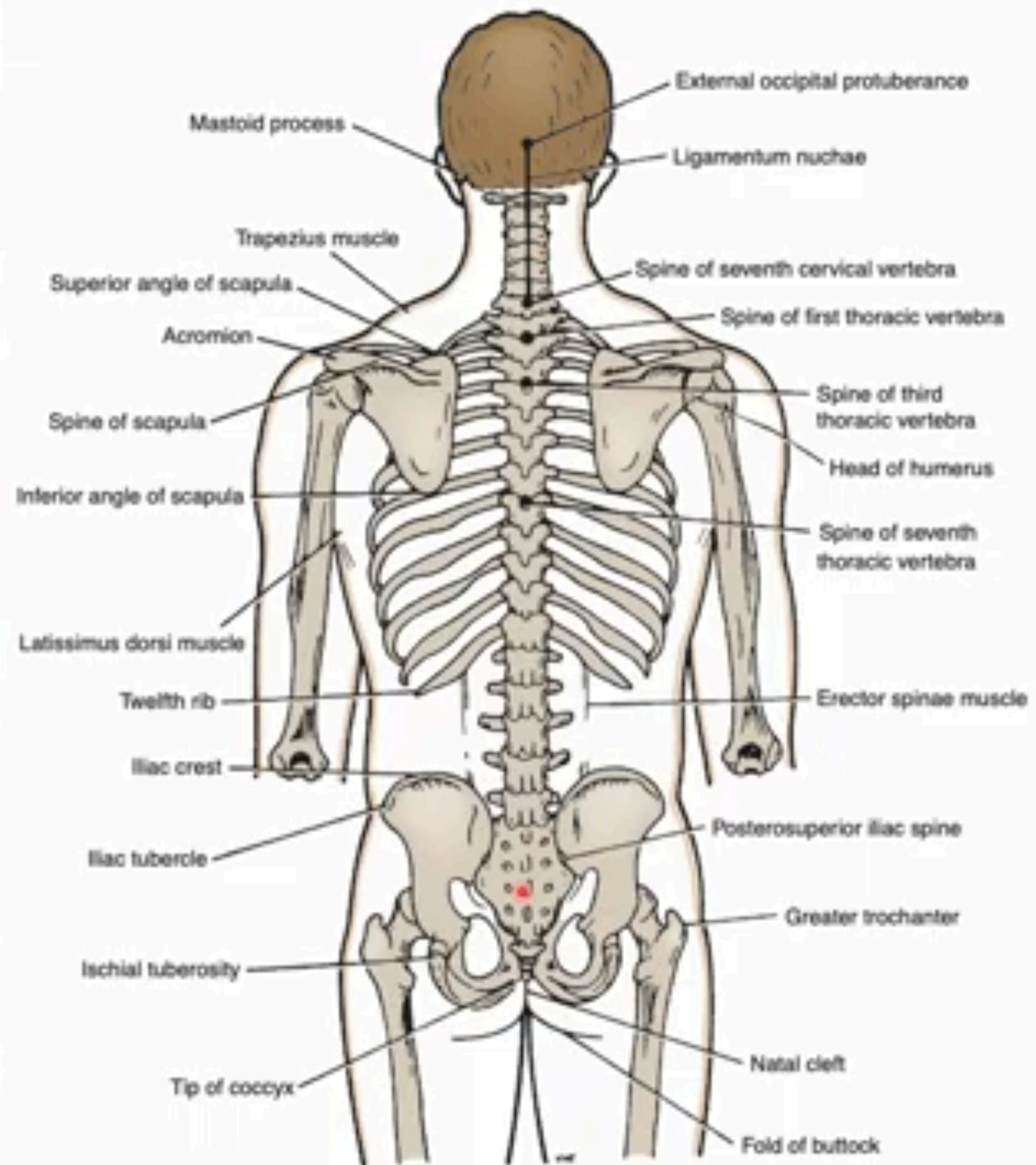
- Introduction of Spinal Cord.
- Lumbar Puncture (Spinal Tap).
- Curvatuers of vertiberal colum deferate end of spinal cord between adult and infant and young child.
- Function of spinal cord.
- Spinal cord has:
 - Internal Featuers.
 - External Features.
- Spinal Meninges amd spaces.
- Spinal Segments.
- Internal Featuers.
- The gray matter of the spinal cord.
- The White matter of Spinal Cord
- Transverse sections of the spinal cord at different levels showing the arrangement of the gray matter and white matter.

- Important notes about the white mater of Spinal Cord.
- Spinal Nerves.
- Transverse section of the spinal cord at the mid cervical level showing the general arrangement of the ascending tracts on the right and the descending tracts on the left.
- Functional components of spinal nerve fibers .
- Components and branches of spinal nerves.
- Arterial supply and venous drainage of spinal cord.
- Histological Structure of Nerve.
- Dermatomes.
- Myotomes.
- Main Descending and ascending Main pathway .

Introduction and Review of The vertebral column

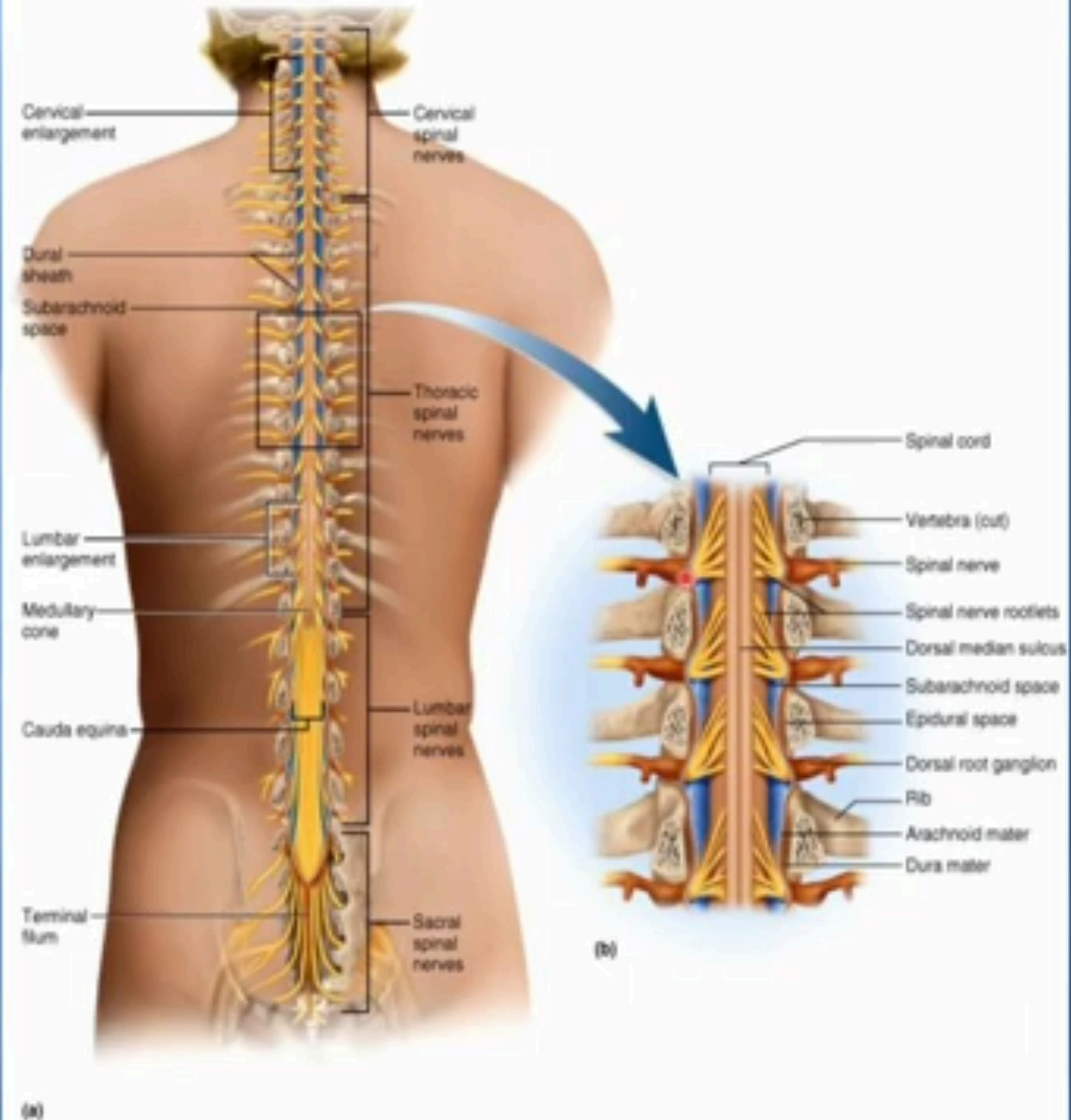
The vertebral column

- It is the **central bony pillar of the body.**
- It **supports the skull, pectoral girdle, upper limbs, and thoracic cage** and, by way of the **pelvic girdle,**
- It **transmits body weight** to the **lower limbs.**



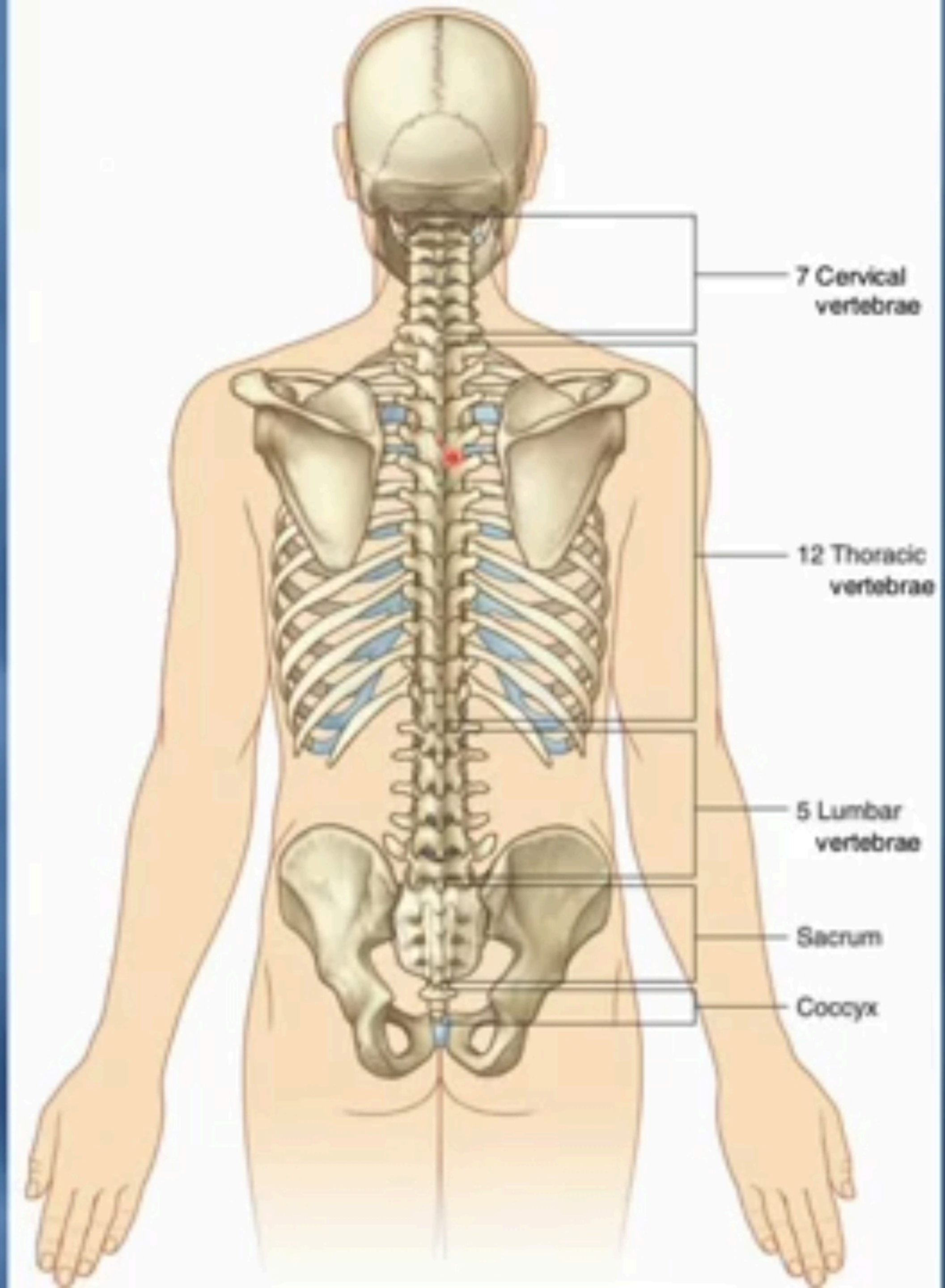
The vertebral column

- Within its **cavity** lie the **spinal cord**, the **roots of the spinal nerves**, and the **covering meninges**, to which the vertebral column gives great **protection**.

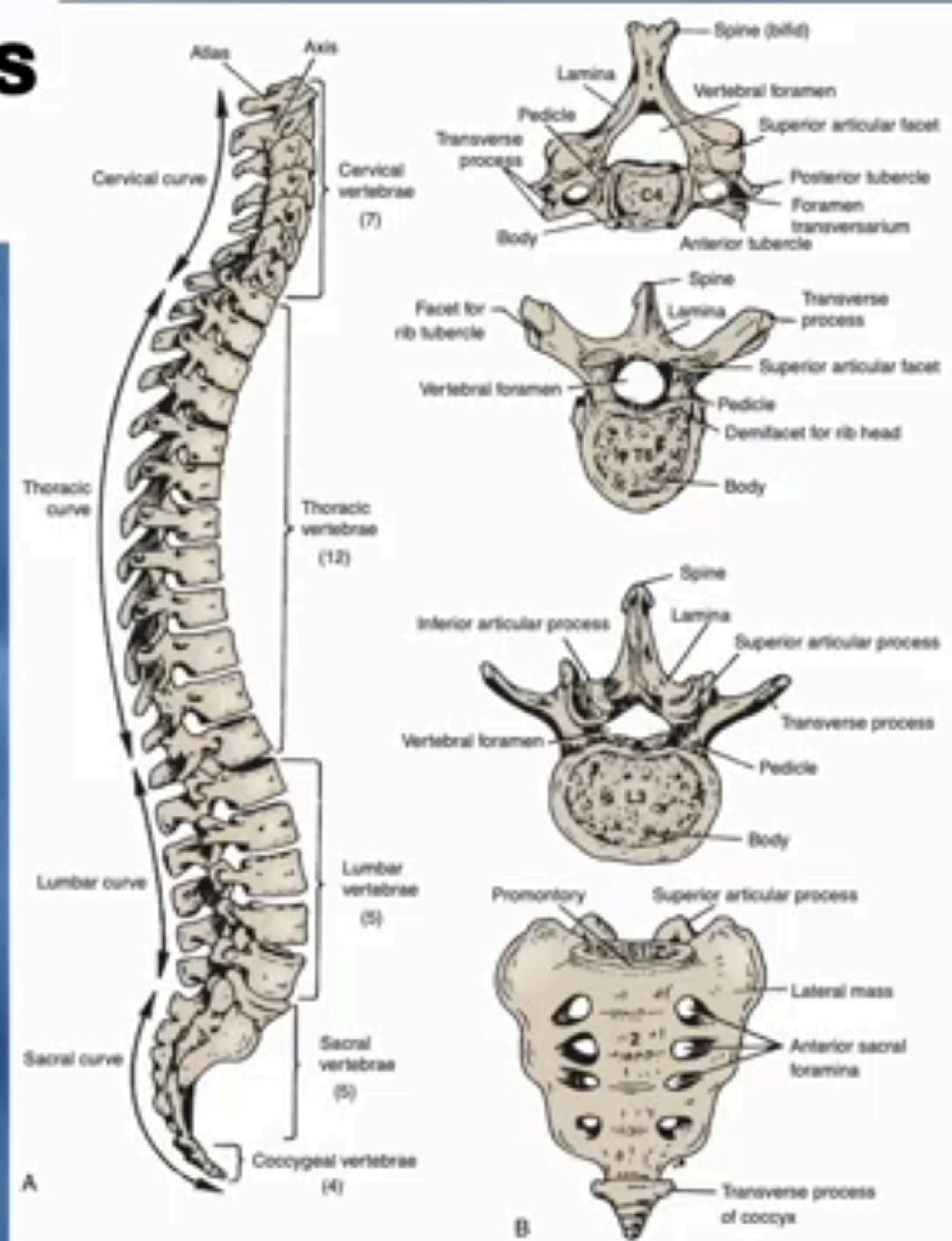
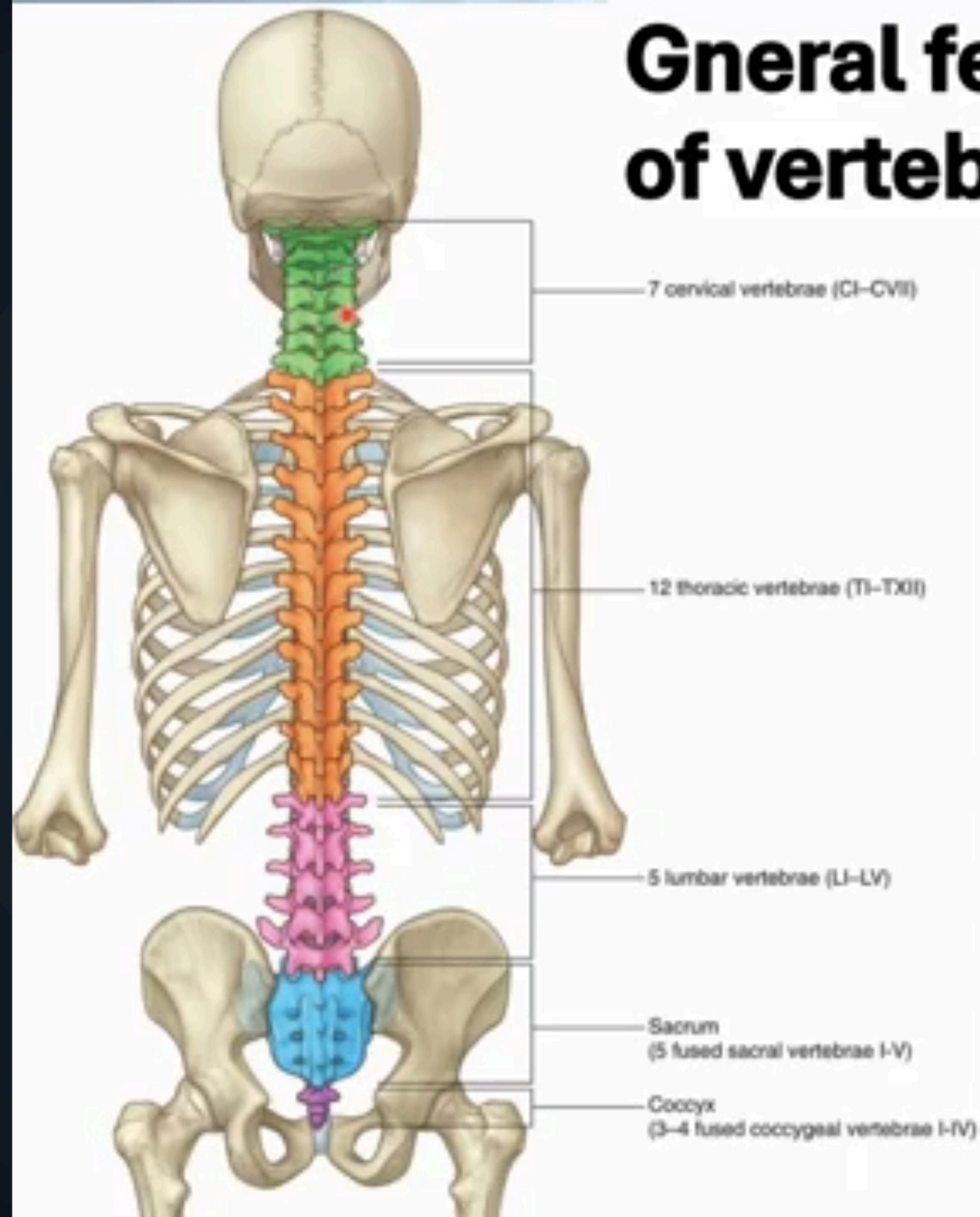


Vertebrae

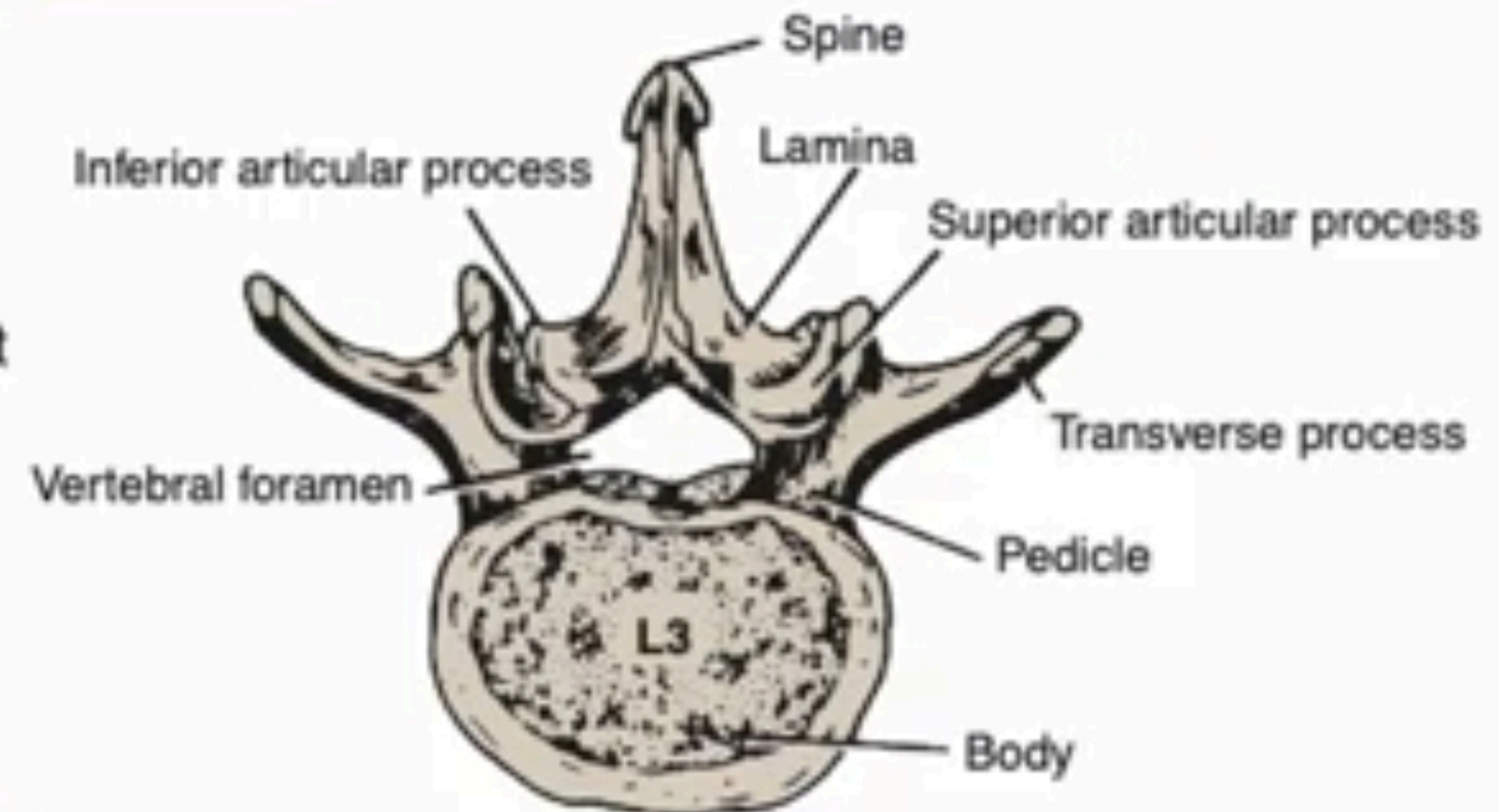
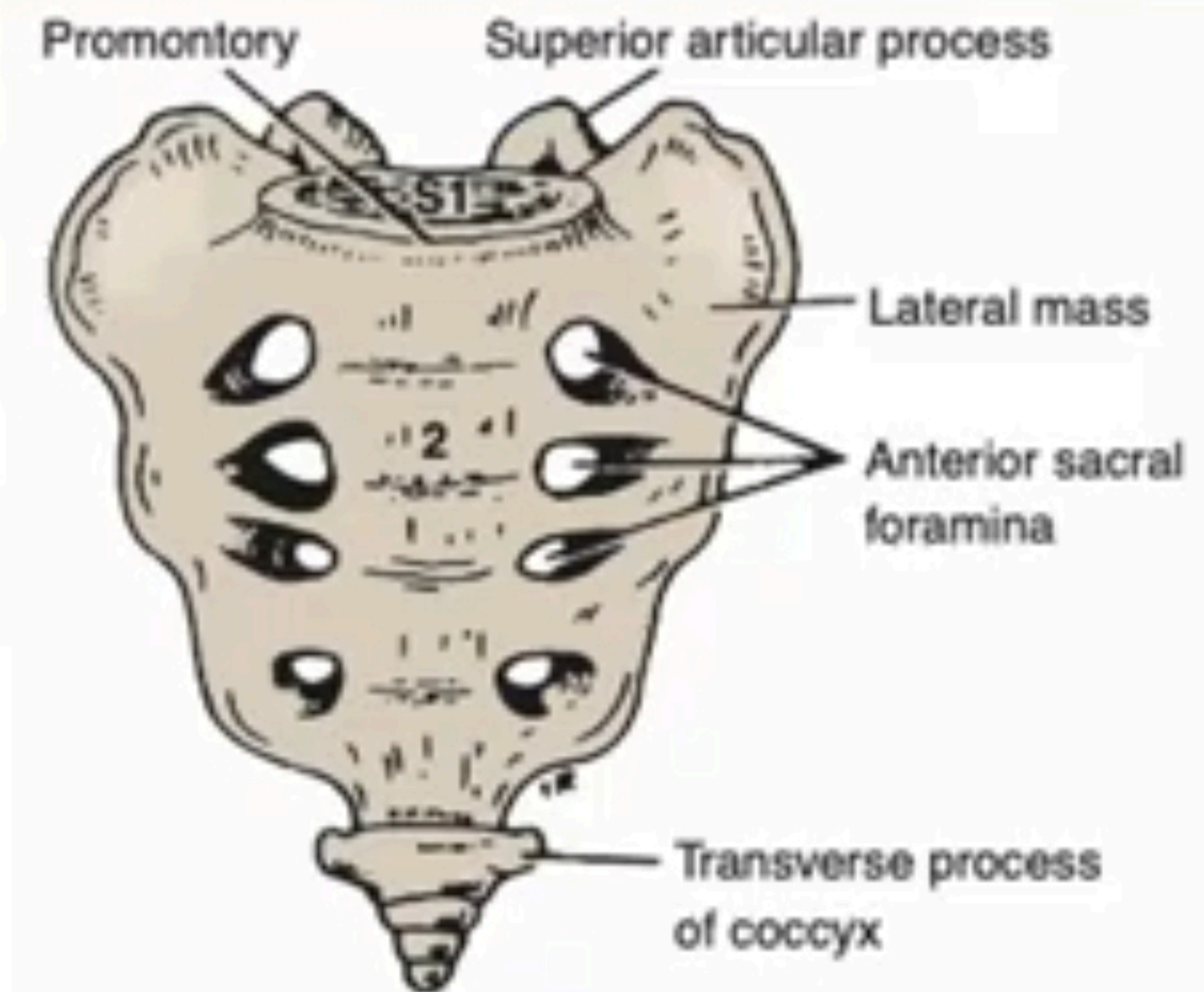
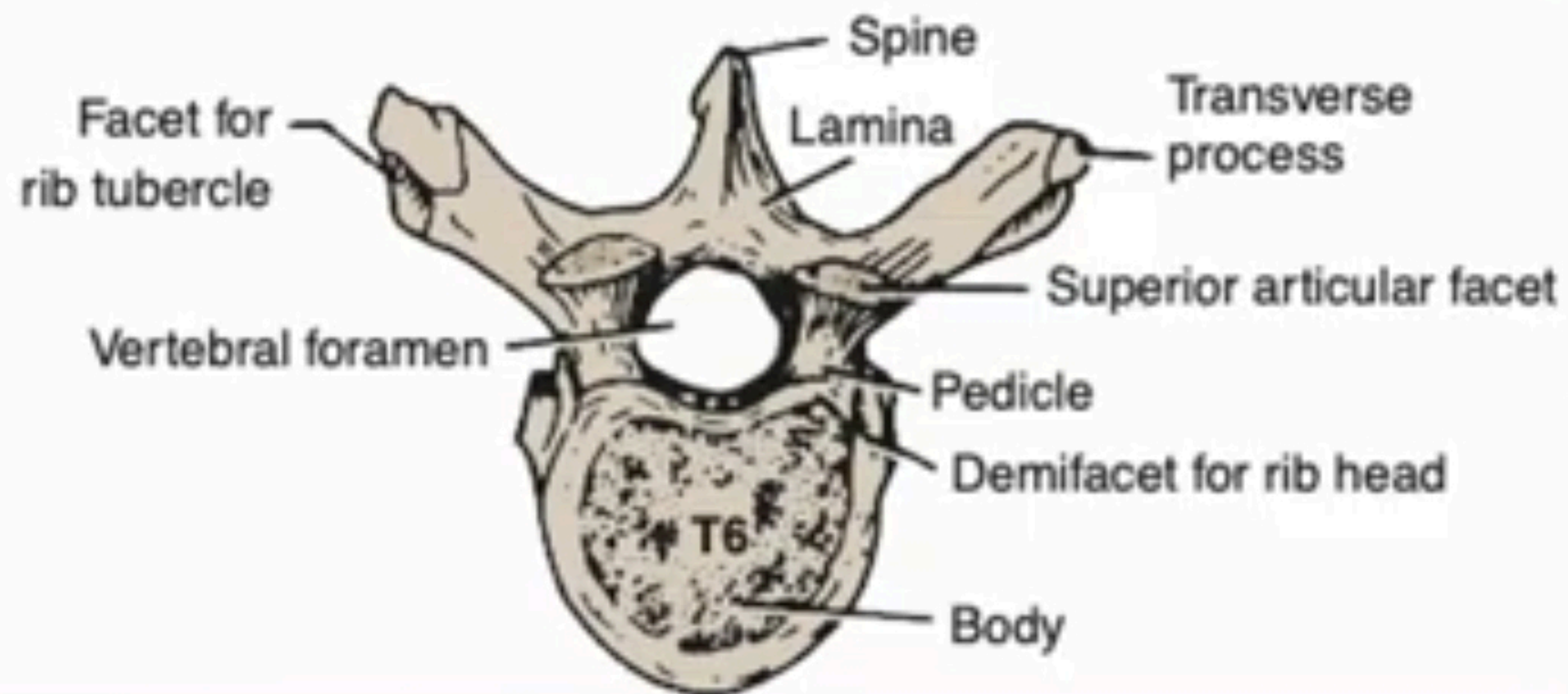
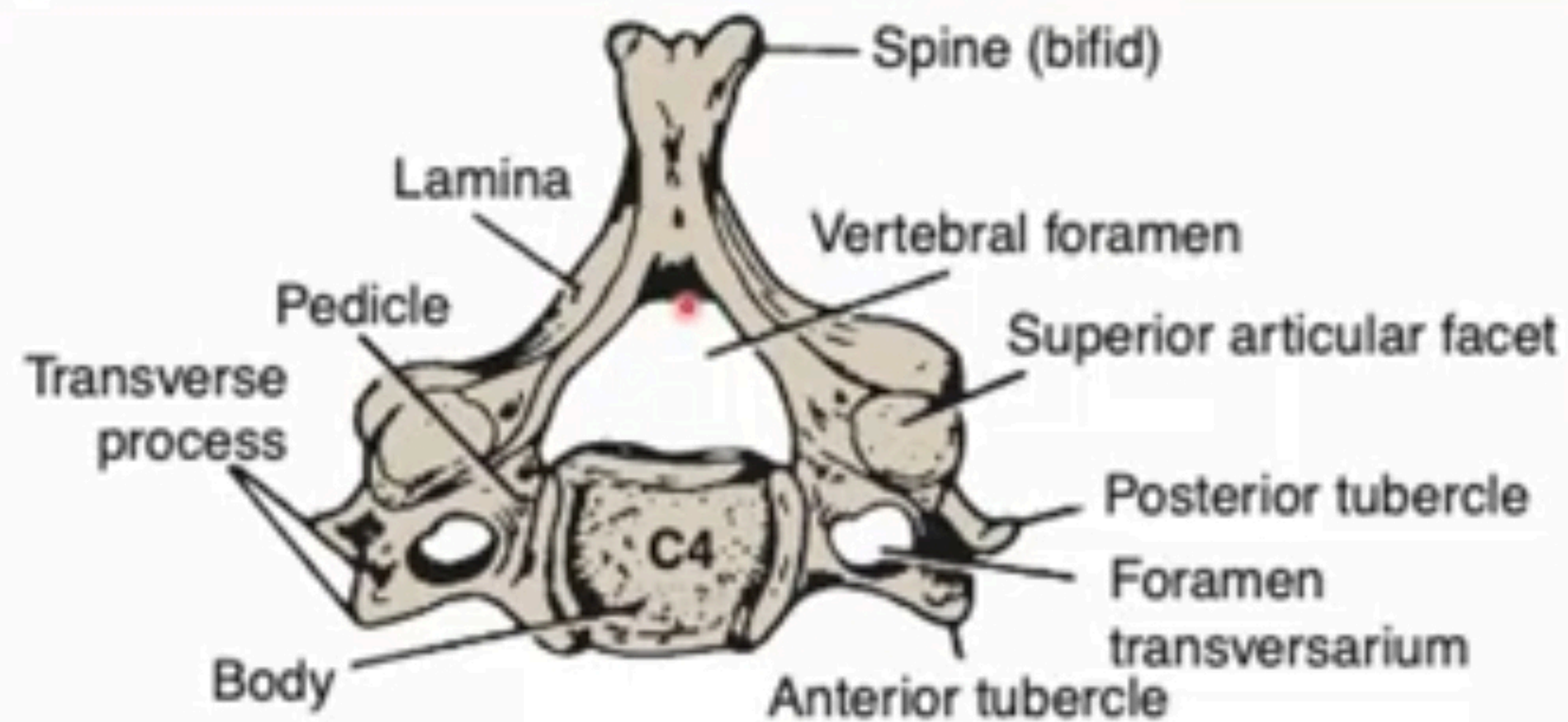
- ❖ There are **approximately 33 vertebrae**, which
- ❖ They are subdivided into **five groups** based on **morphology** and **location**:
 - 7 cervical vertebrae.
 - 12 thoracic vertebrae.
 - 5 lumbar vertebrae.
 - Sacrum (5 fused).
 - Coccyx (3 or 4 fused).



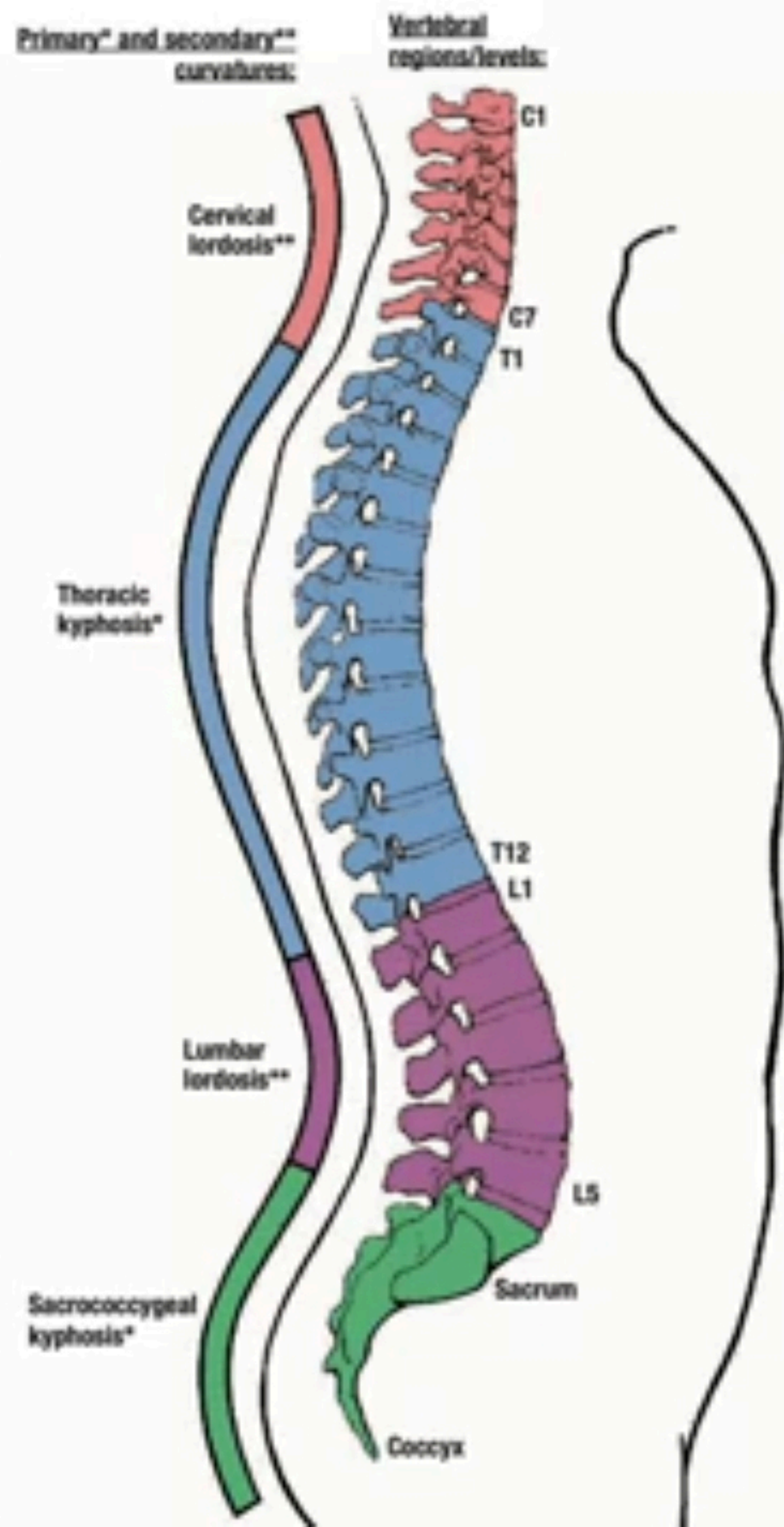
Gneral featuers of vertebrae



Gneral featuers of vertebrae

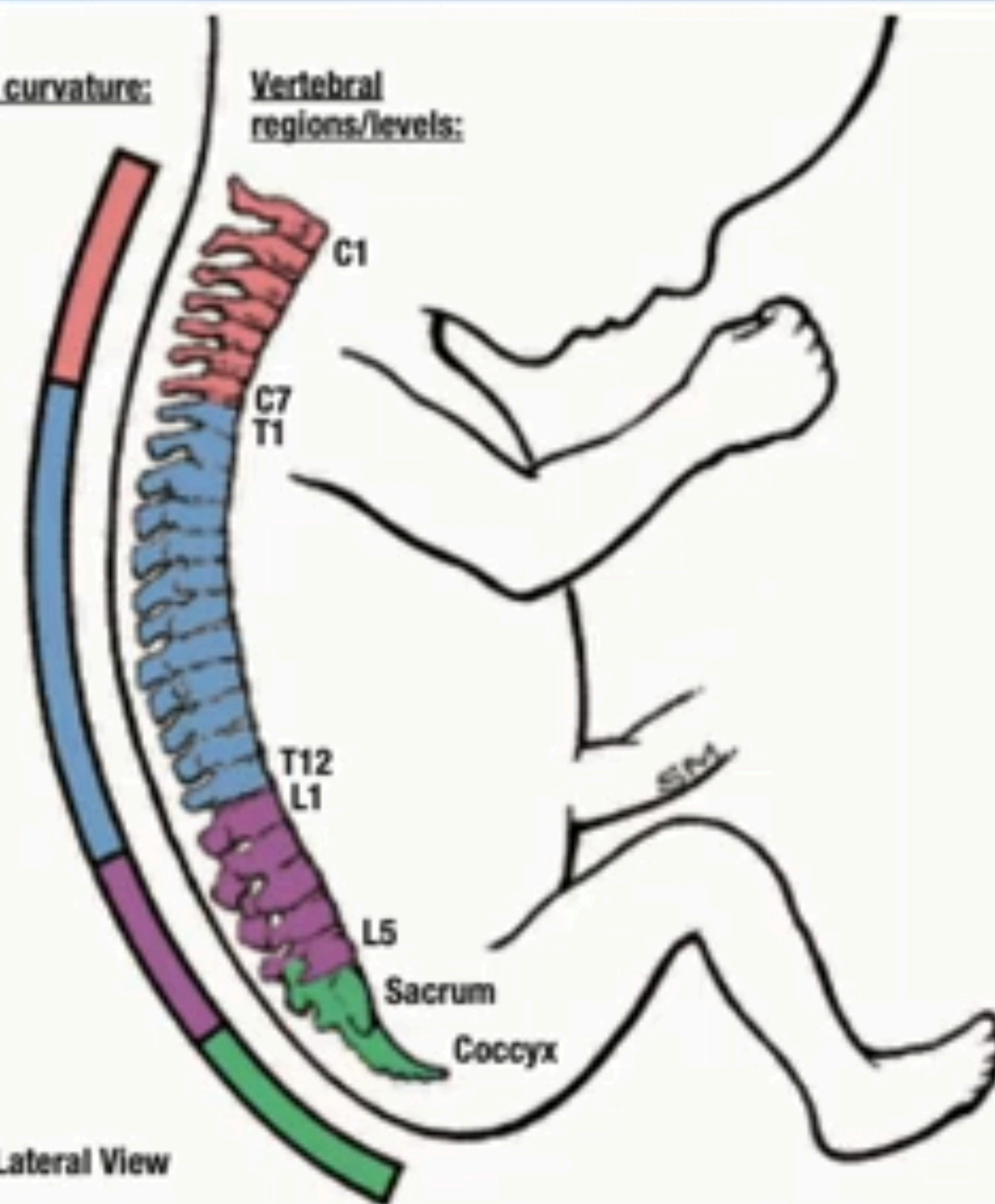


Curvatures of vertebral column



Primary curvature:

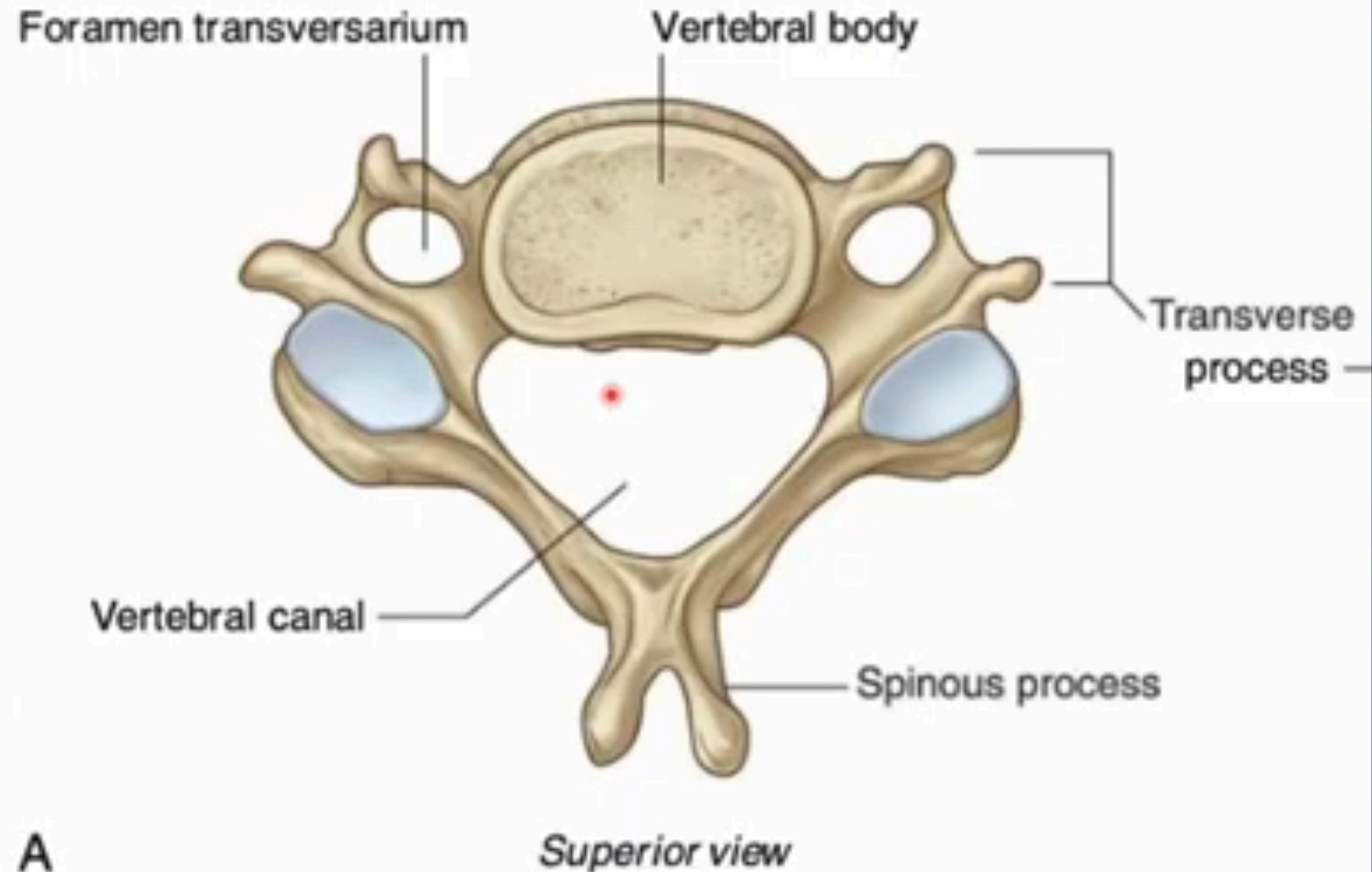
Vertebral regions/levels:



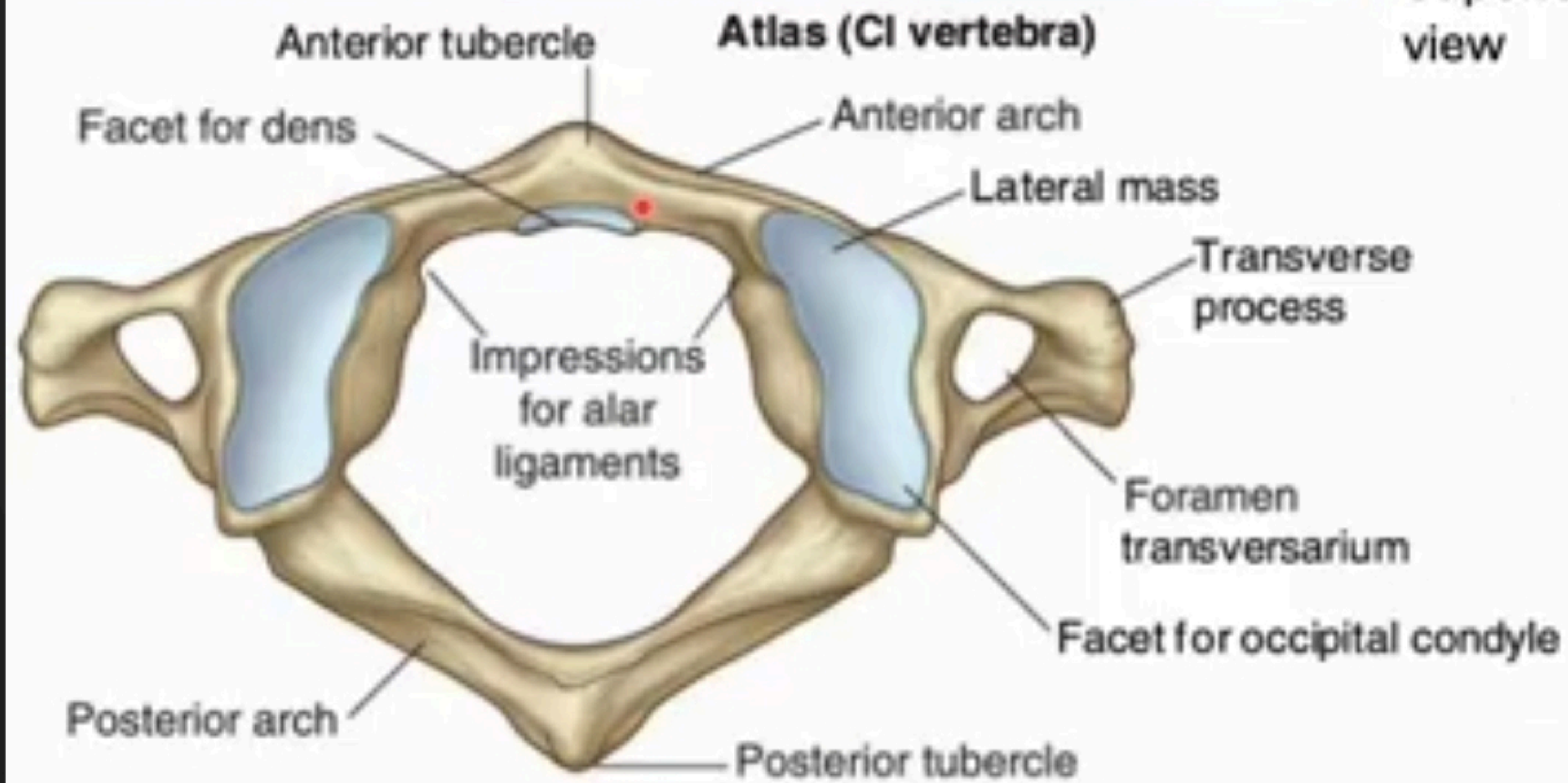
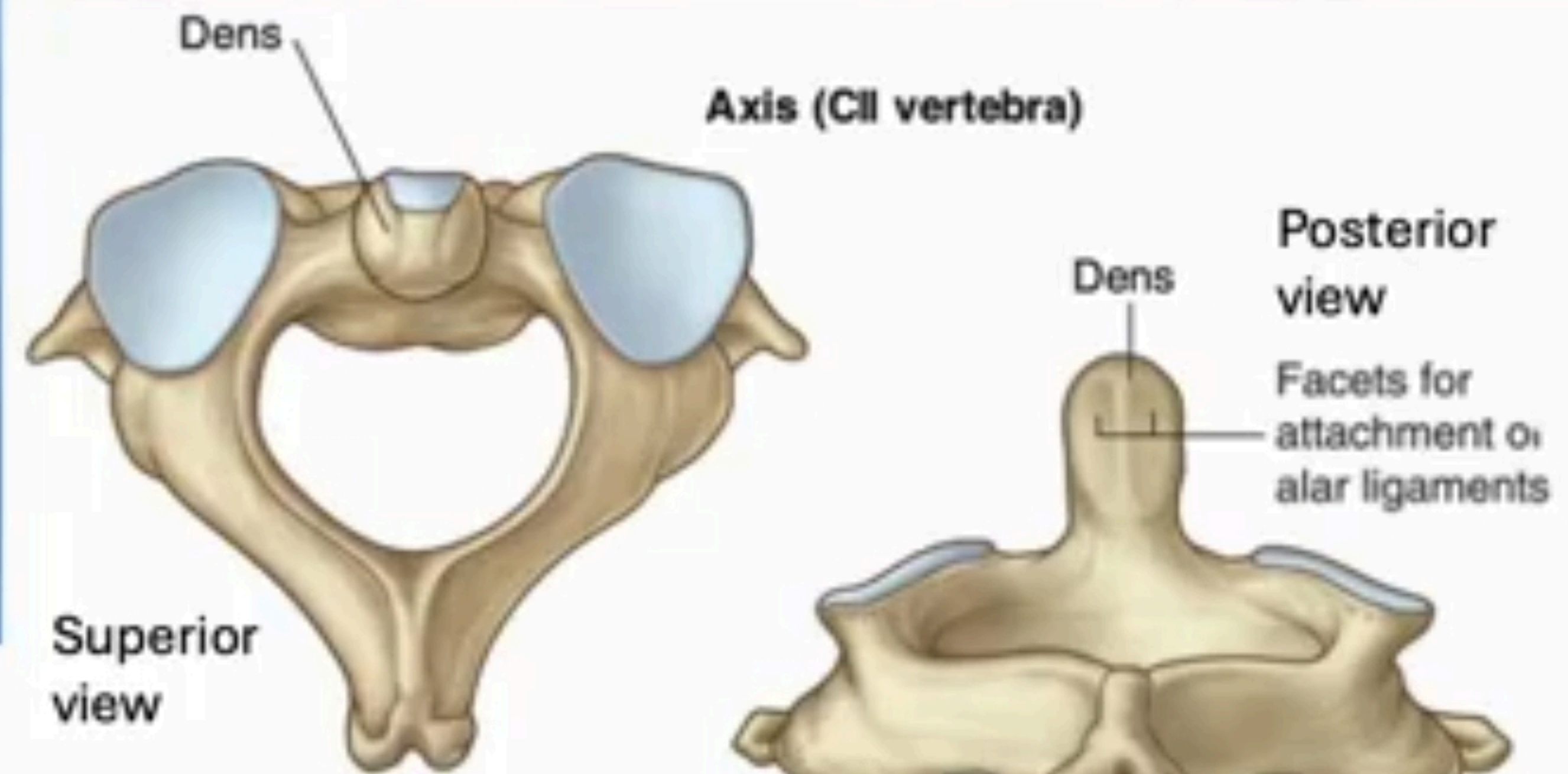
A. Lateral View

Cervical Vertebrae

- ✓ The **seven** cervical vertebrae
- ✓ It is between the skull and thorax.
- ✓ They are characterized mainly by their **small size** and the presence of a **foramen** in each **transverse process**.



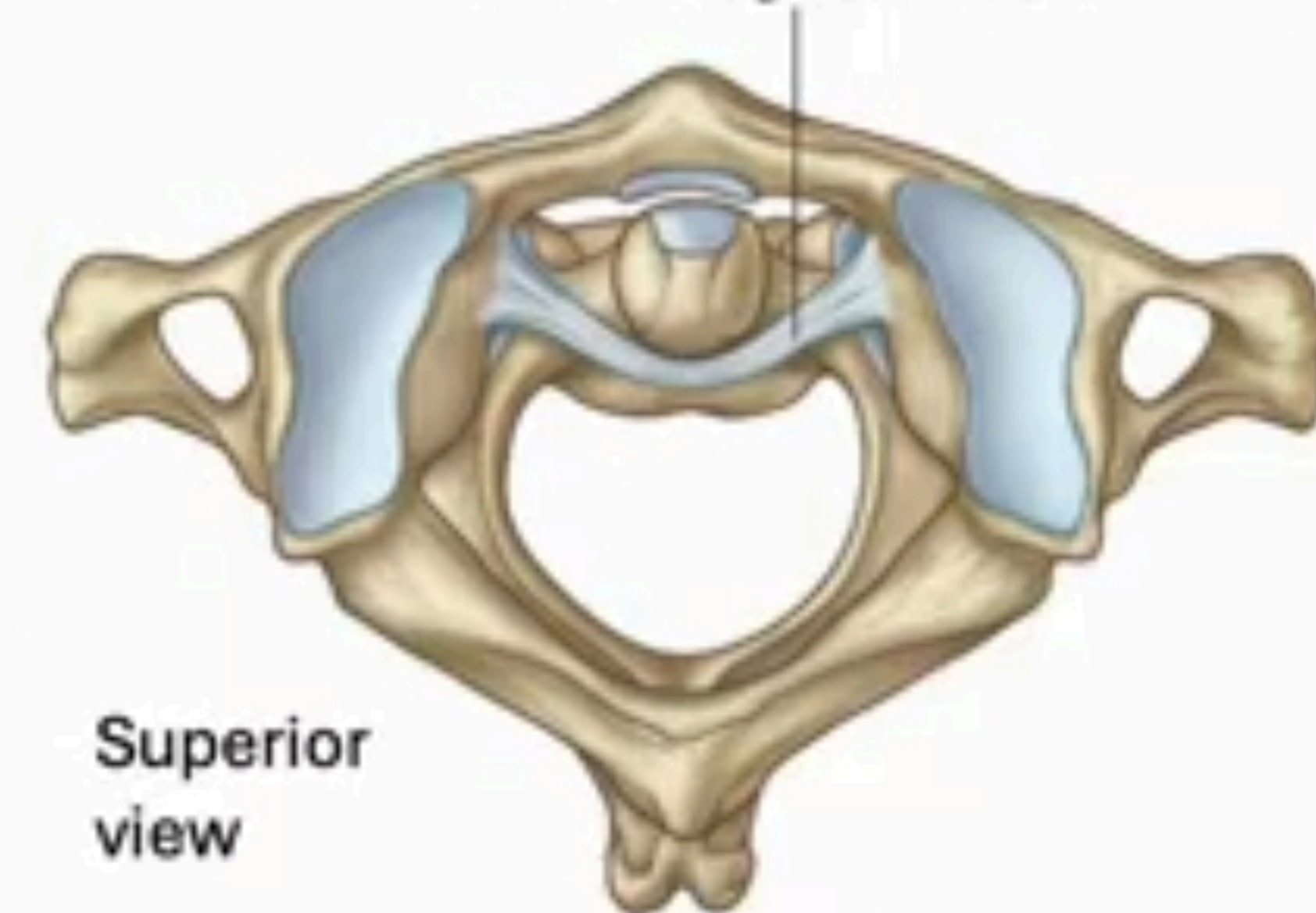
First and Second Cervical Vertebrae



Superior view

Atlas (CI vertebra) and Axis (CII vertebra)

Transverse ligament of atlas

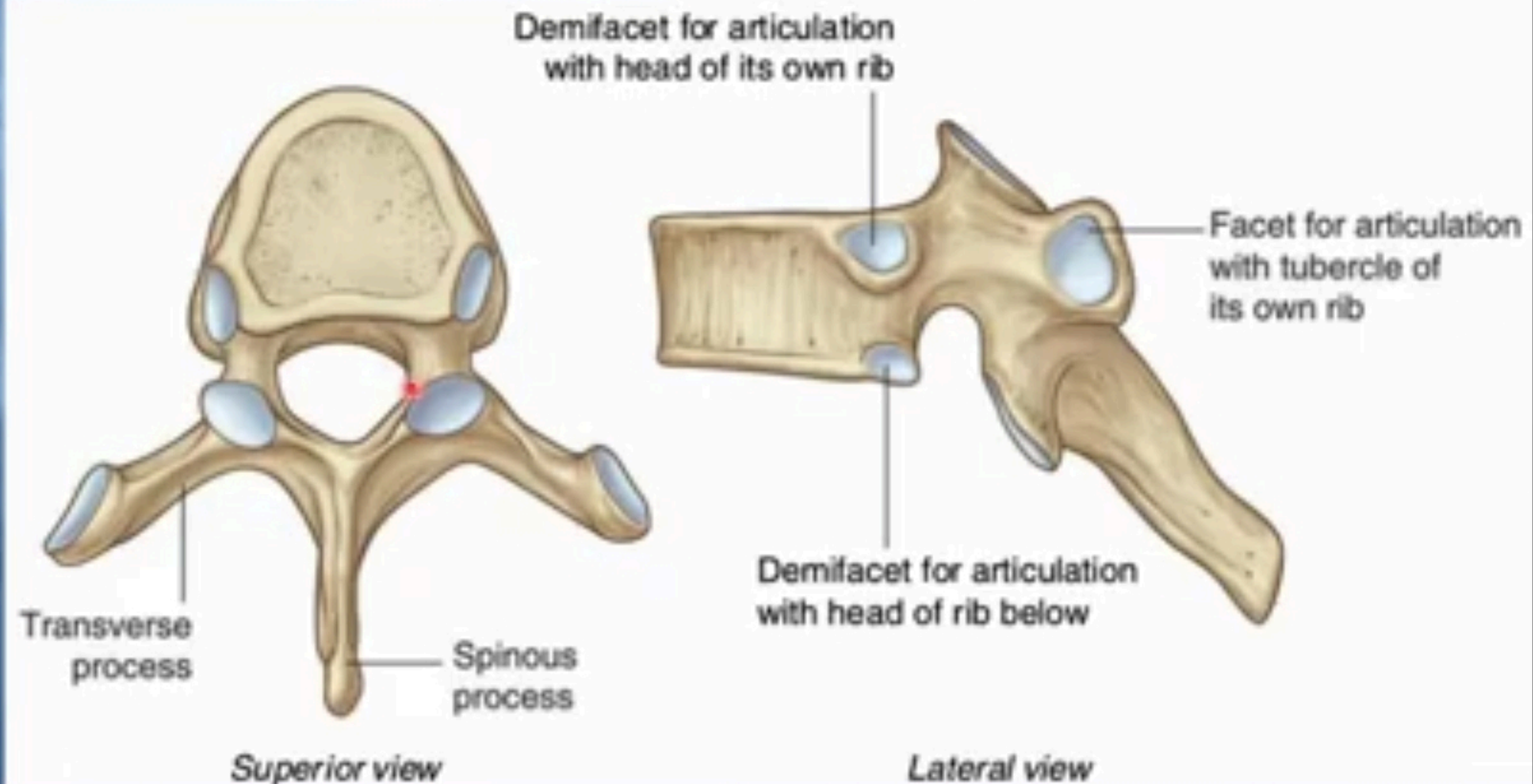


The Thoracic Vertebrae

- The **12 thoracic vertebrae** are characterized by **their articulated ribs**.
- «ll vertebrae have rib elements.
- the ribs are separate bones and articulate via **synovial joints** with the **vertebral bodies** and **transverse processes** of the associated vertebrae.
-

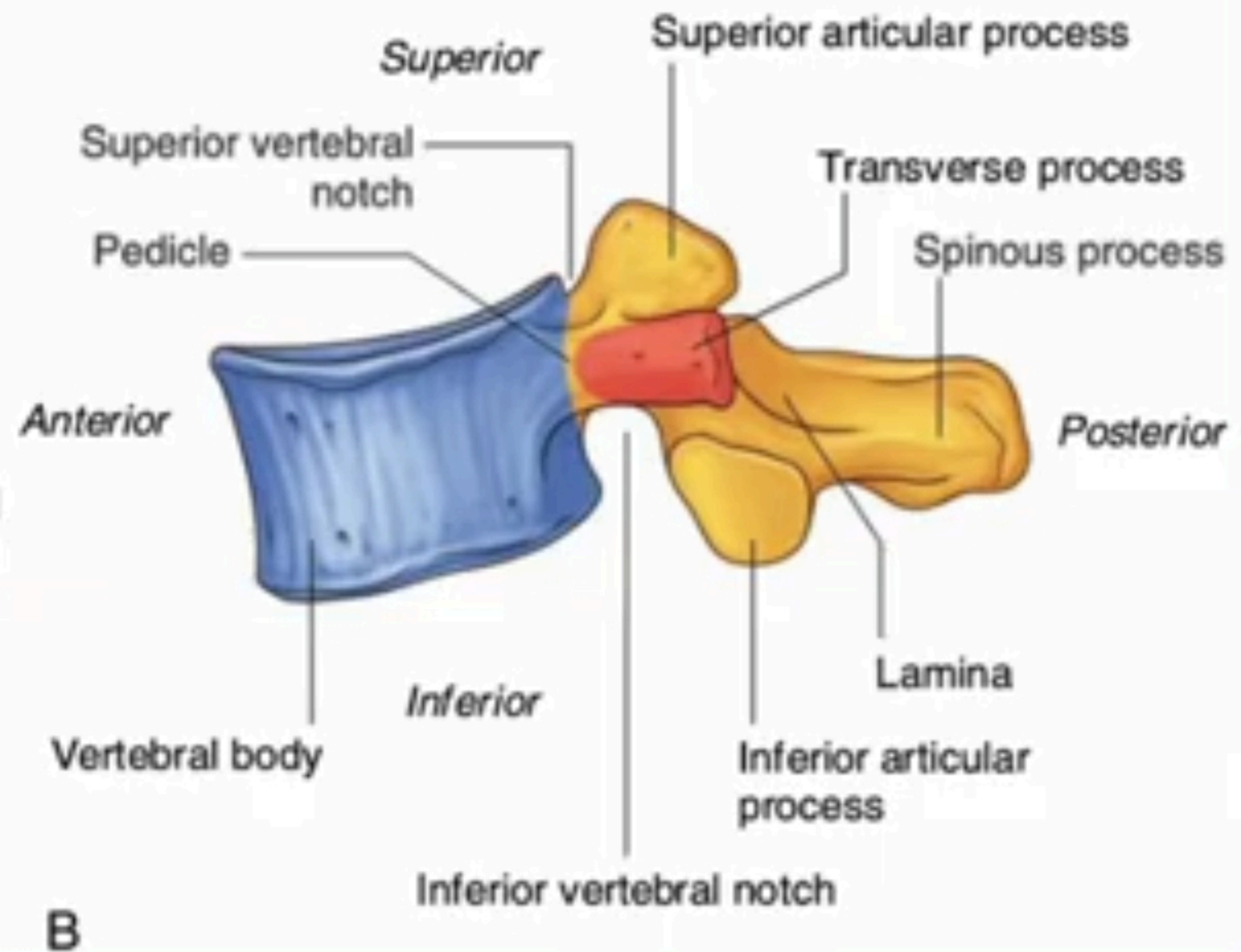
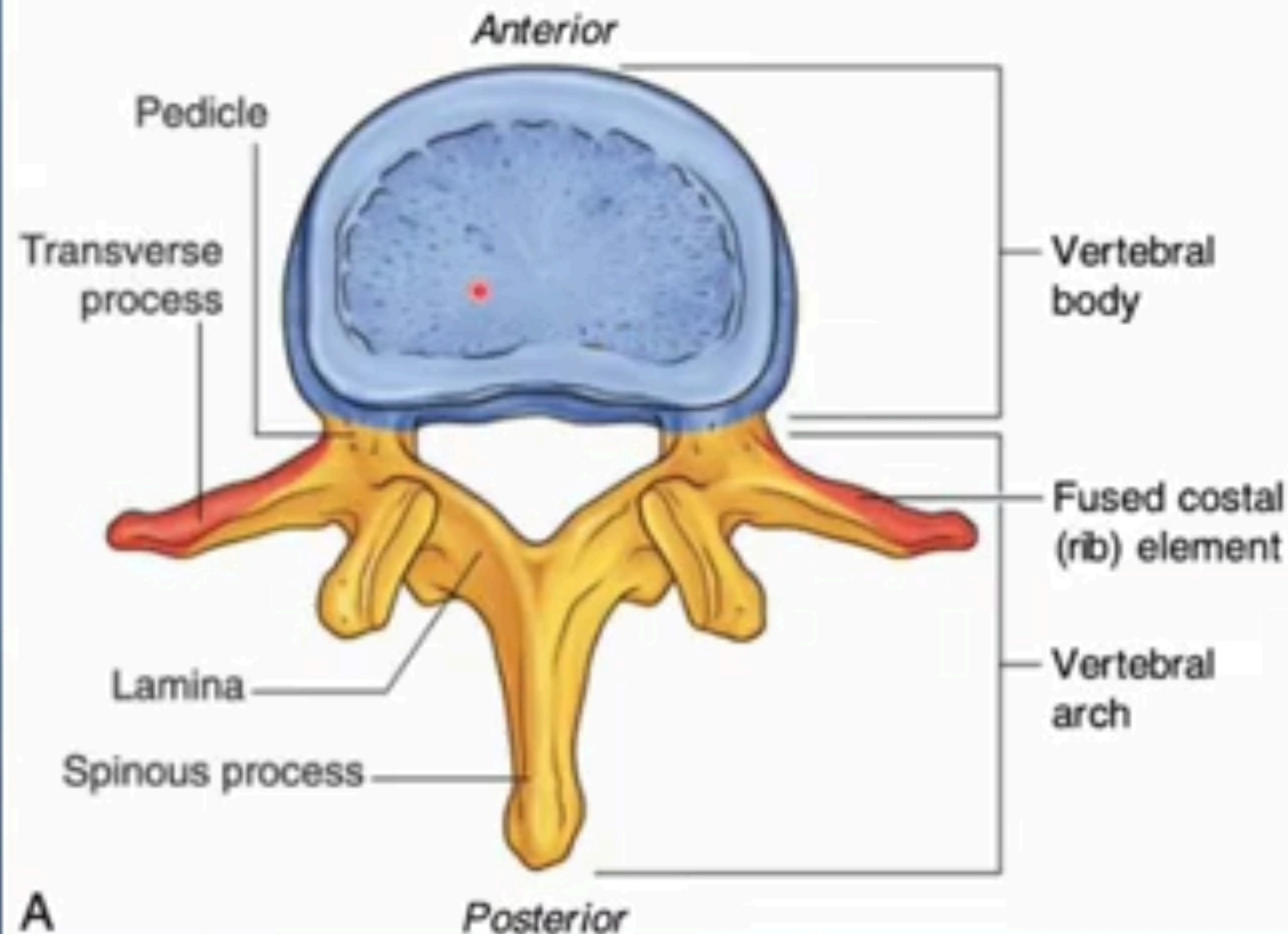


Thoracic vertebra

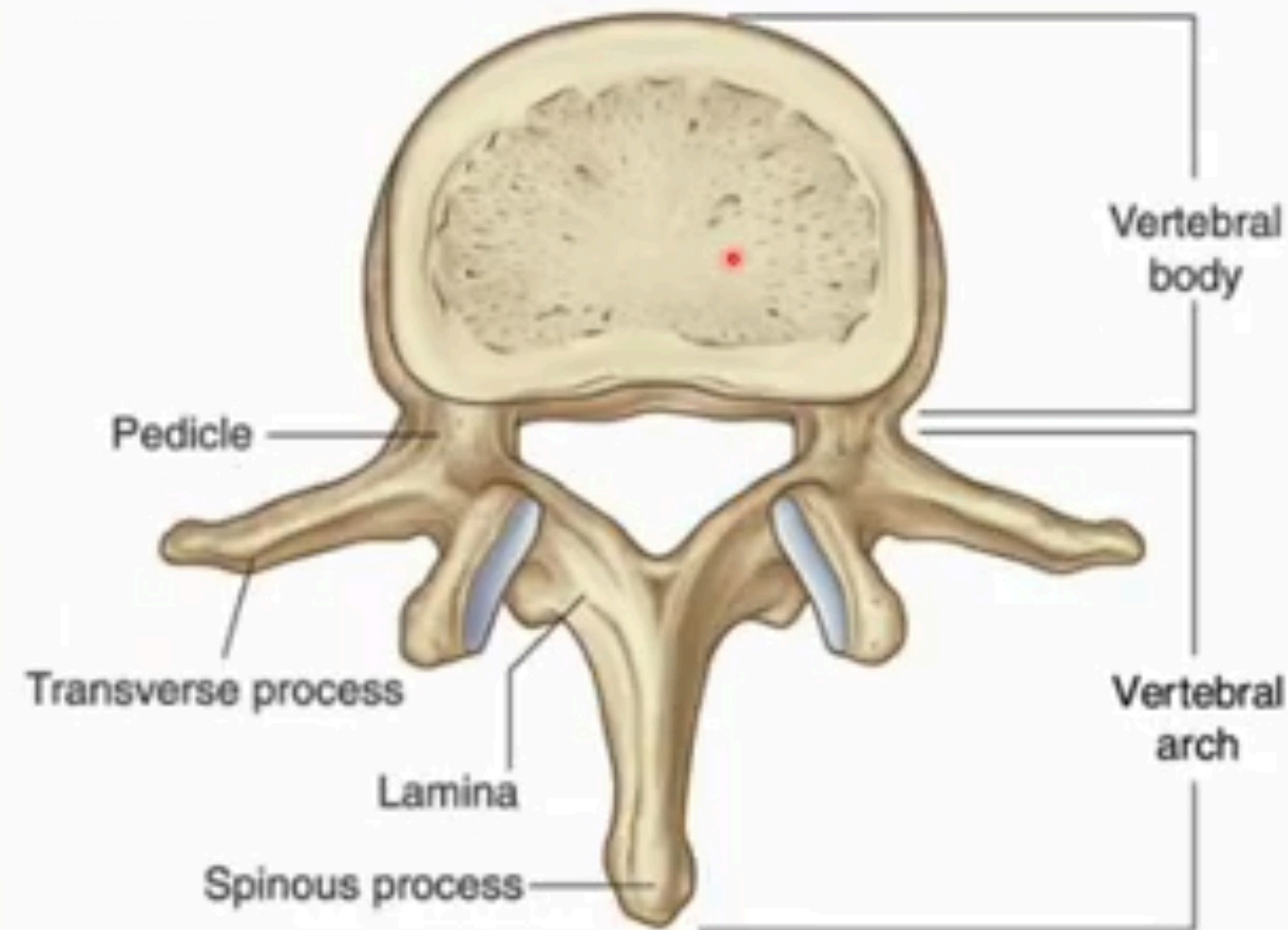


The lumbar vertebrae

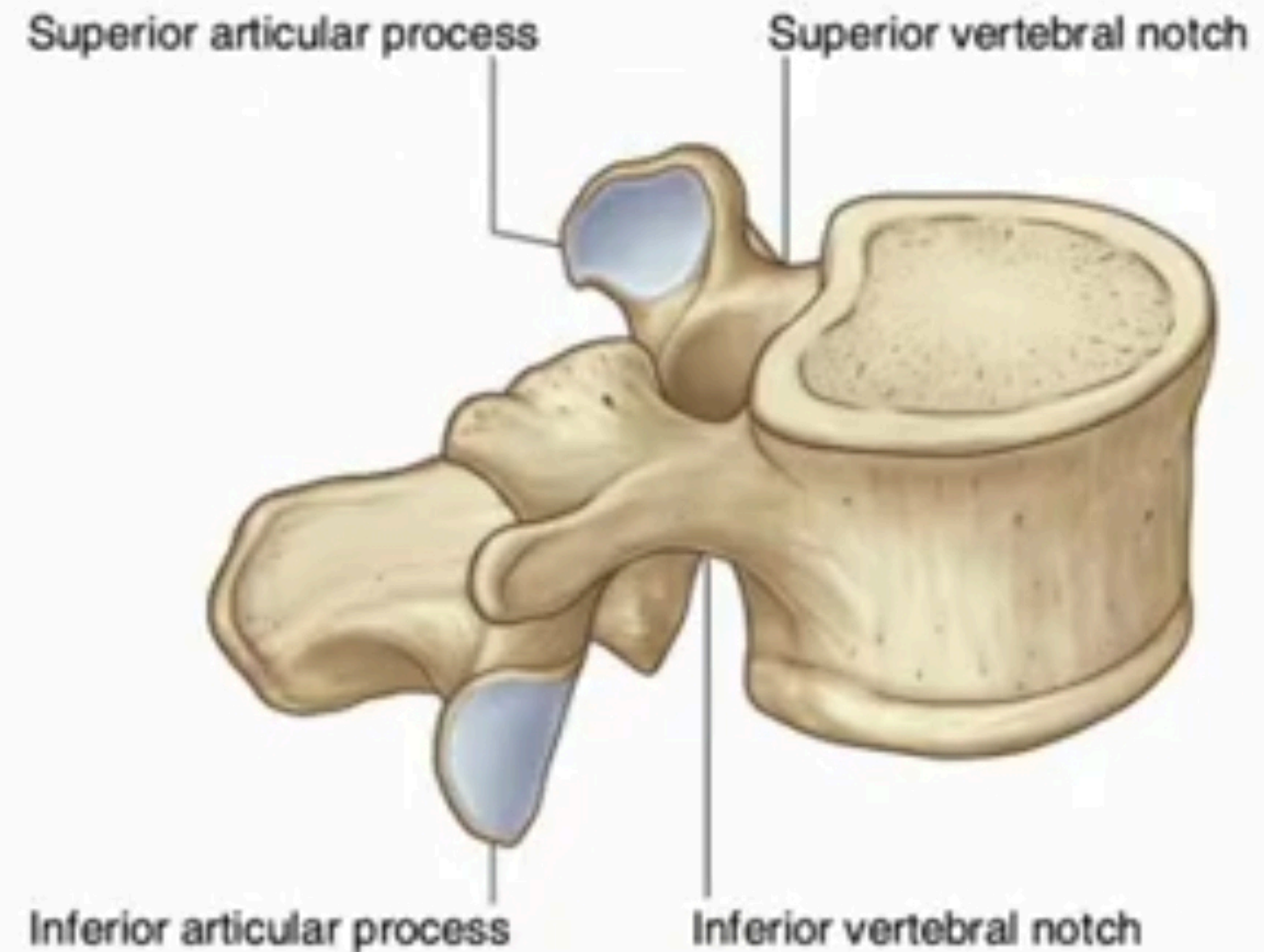
- **Inferior to the thoracic vertebrae** are **five lumbar vertebrae**, which form the **skeletal support** for the **posterior** abdominal wall and are characterized by their **large size**.



The lumbar vertebrae

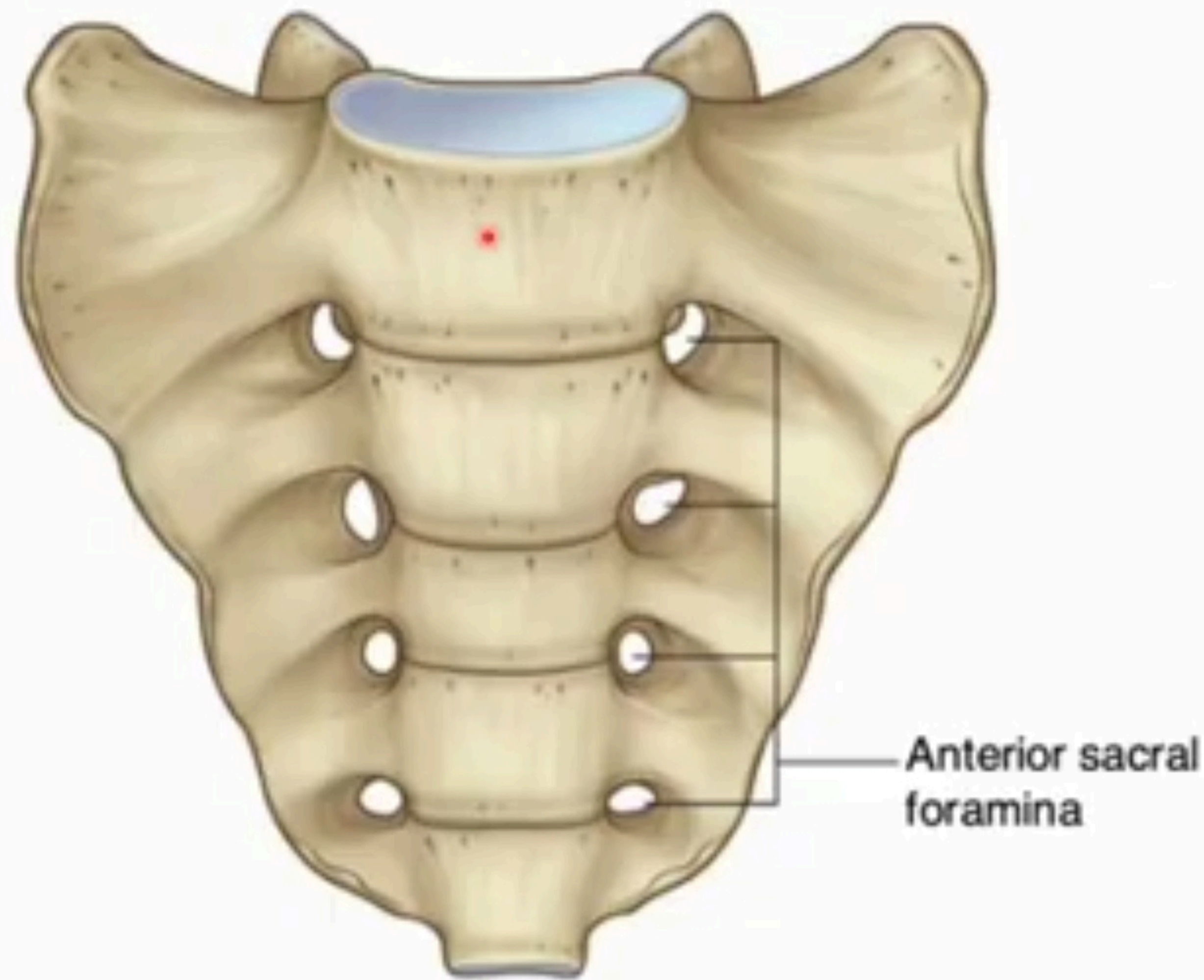


Superior view

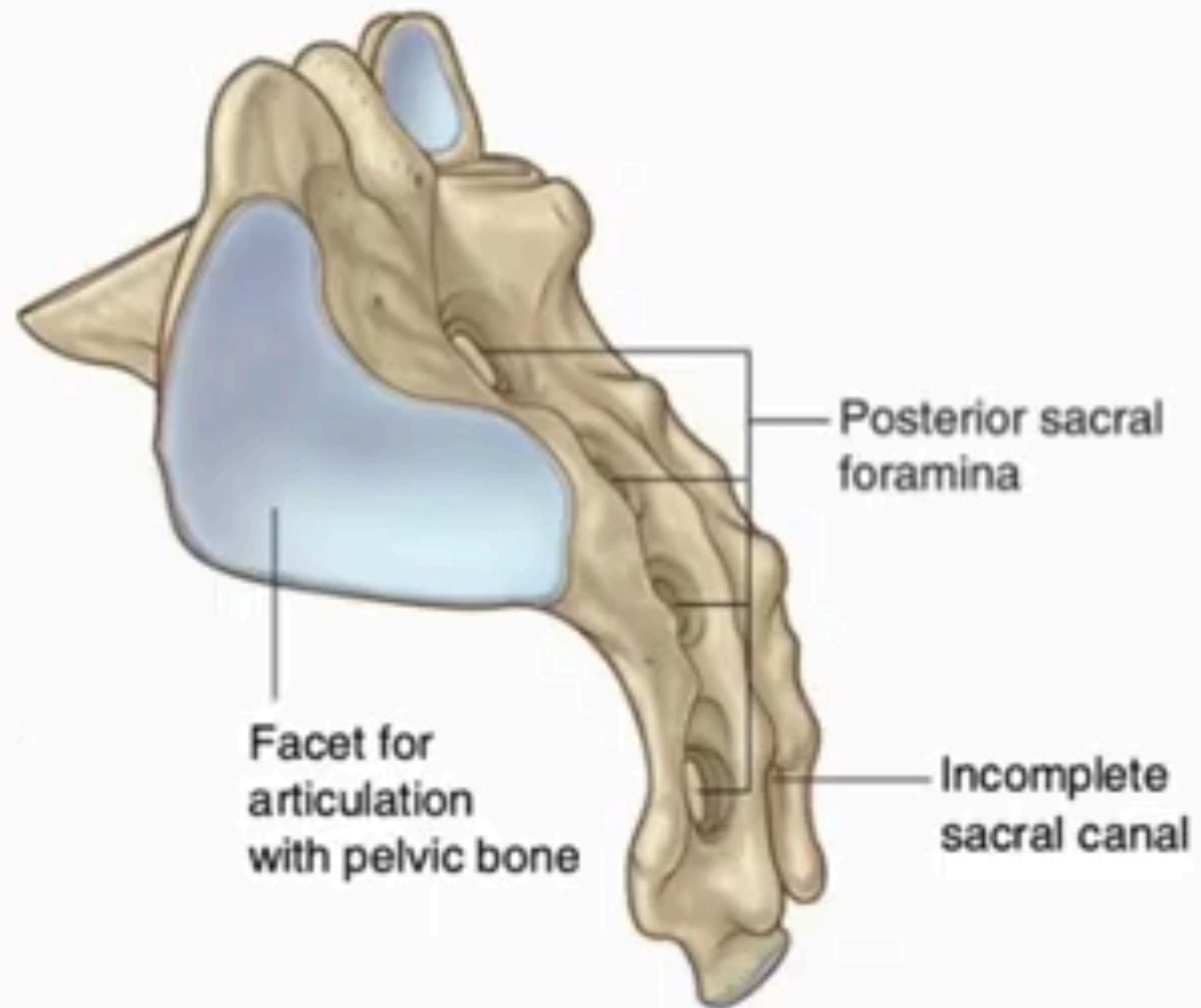


Superolateral oblique view

Sacrum and Coccygeas

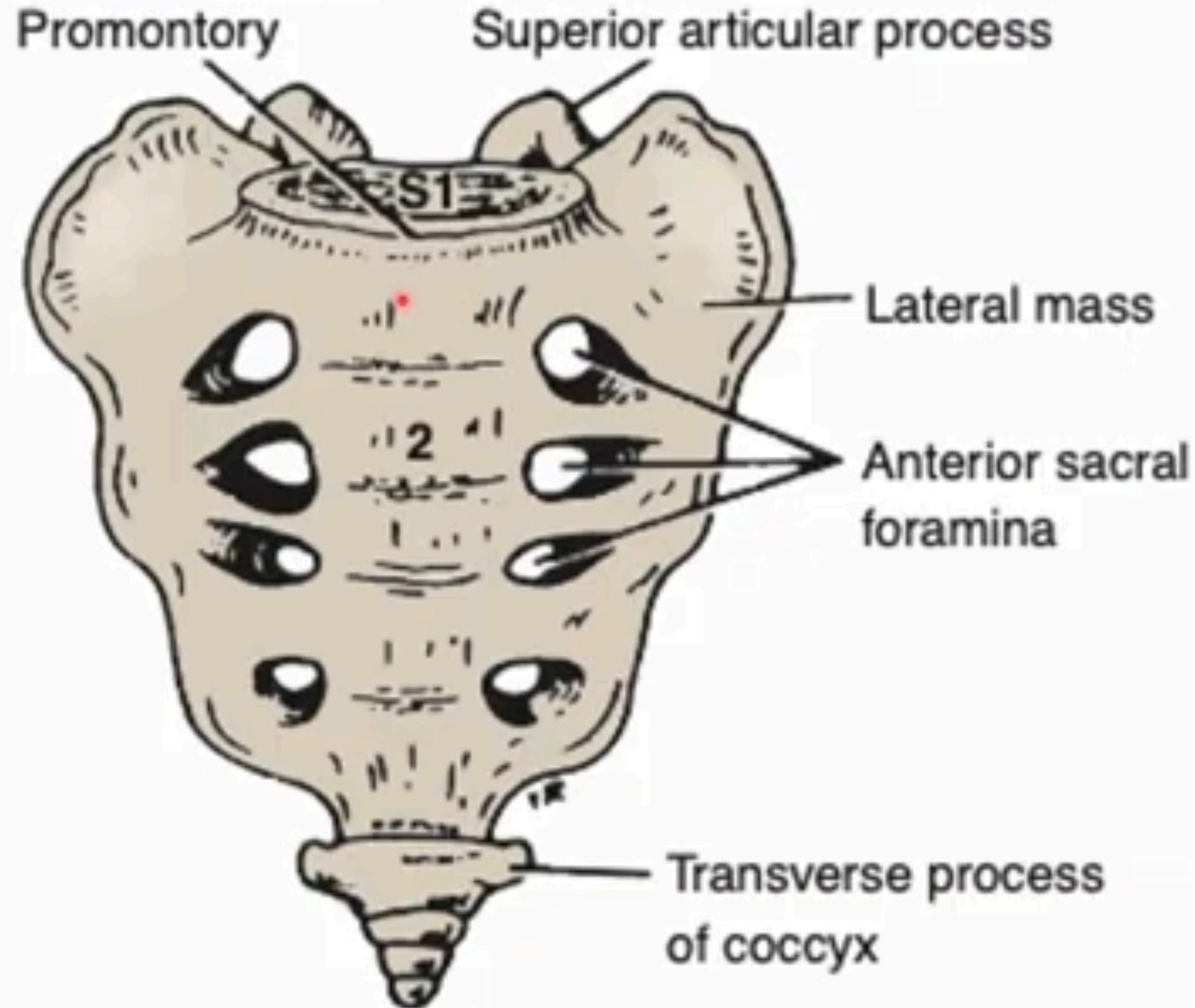


Anterior view



Dorsolateral view

Sacrum and Coccygeas

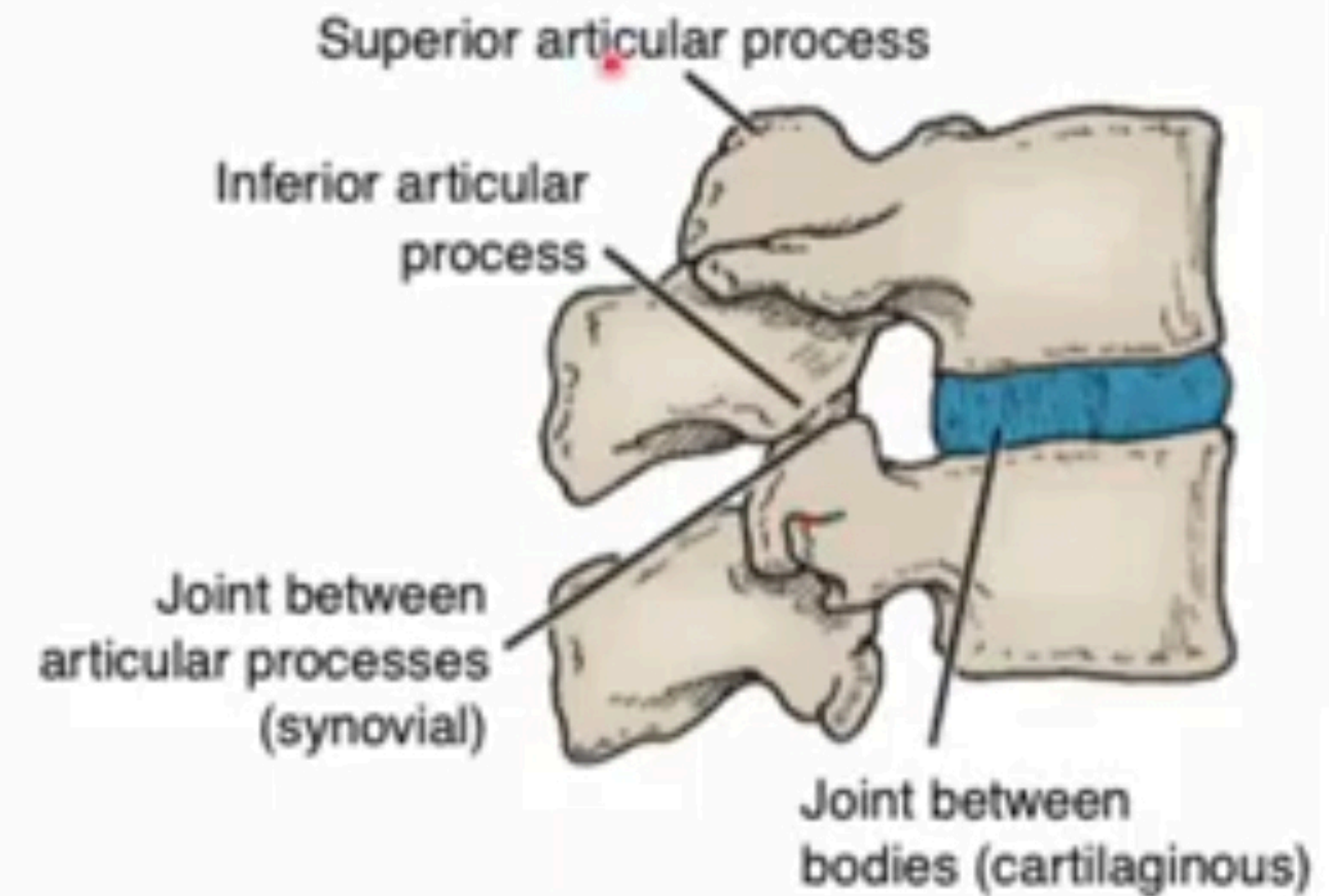
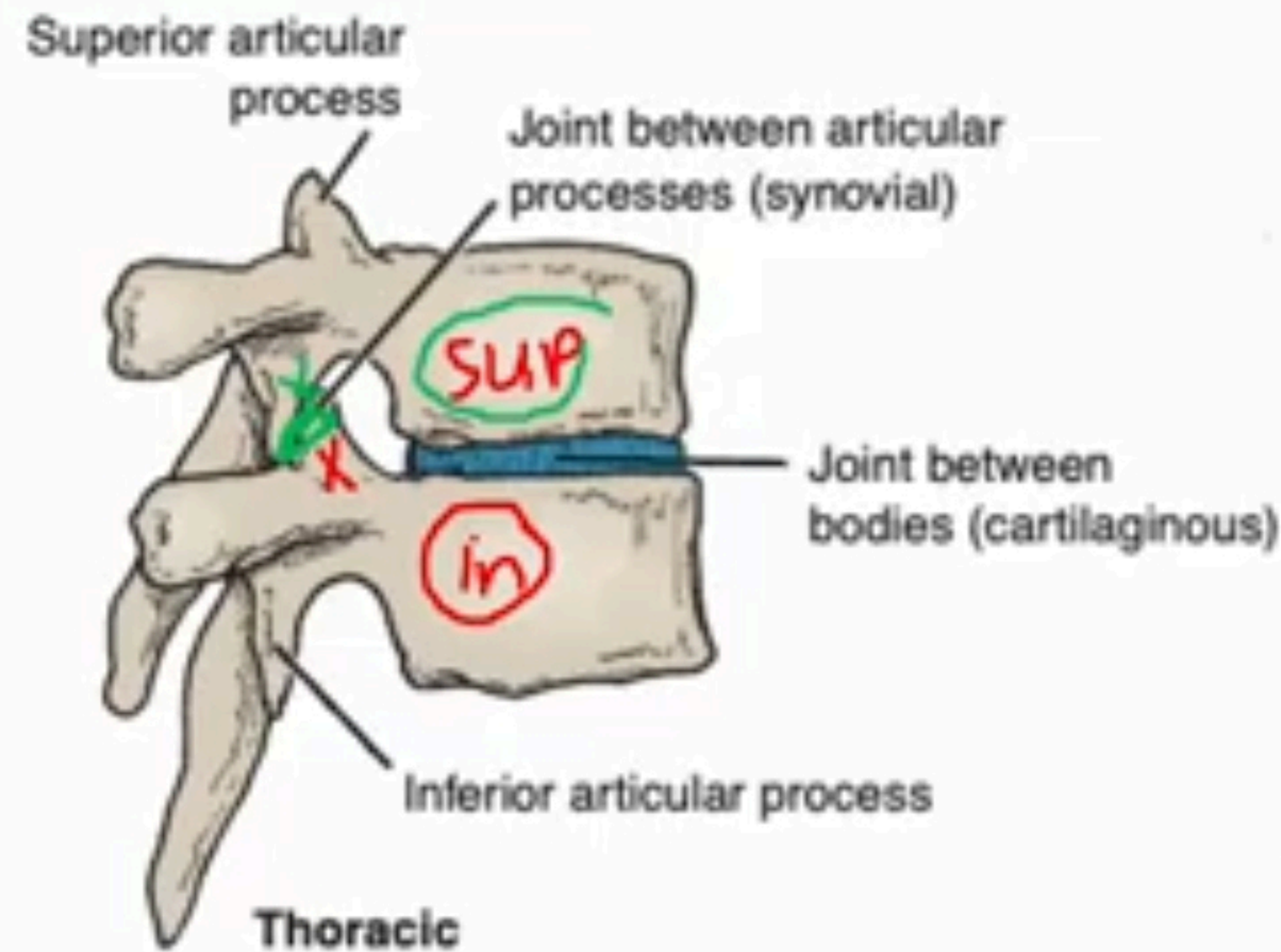
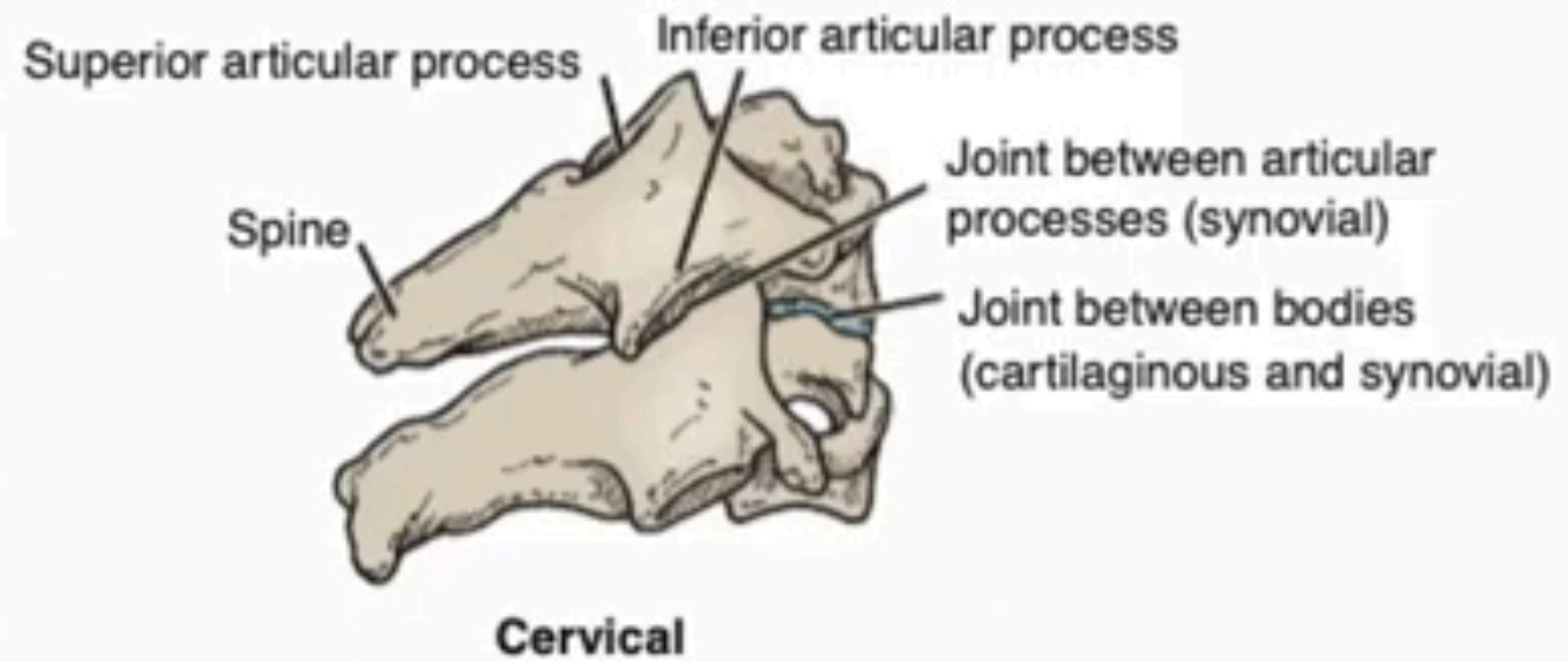


Coccygeal cornu

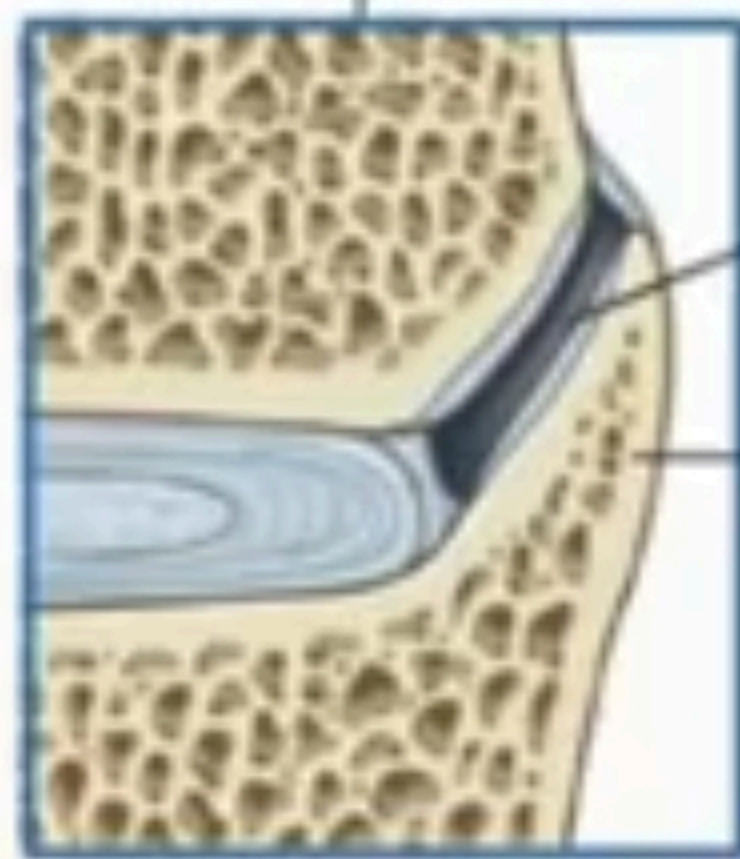
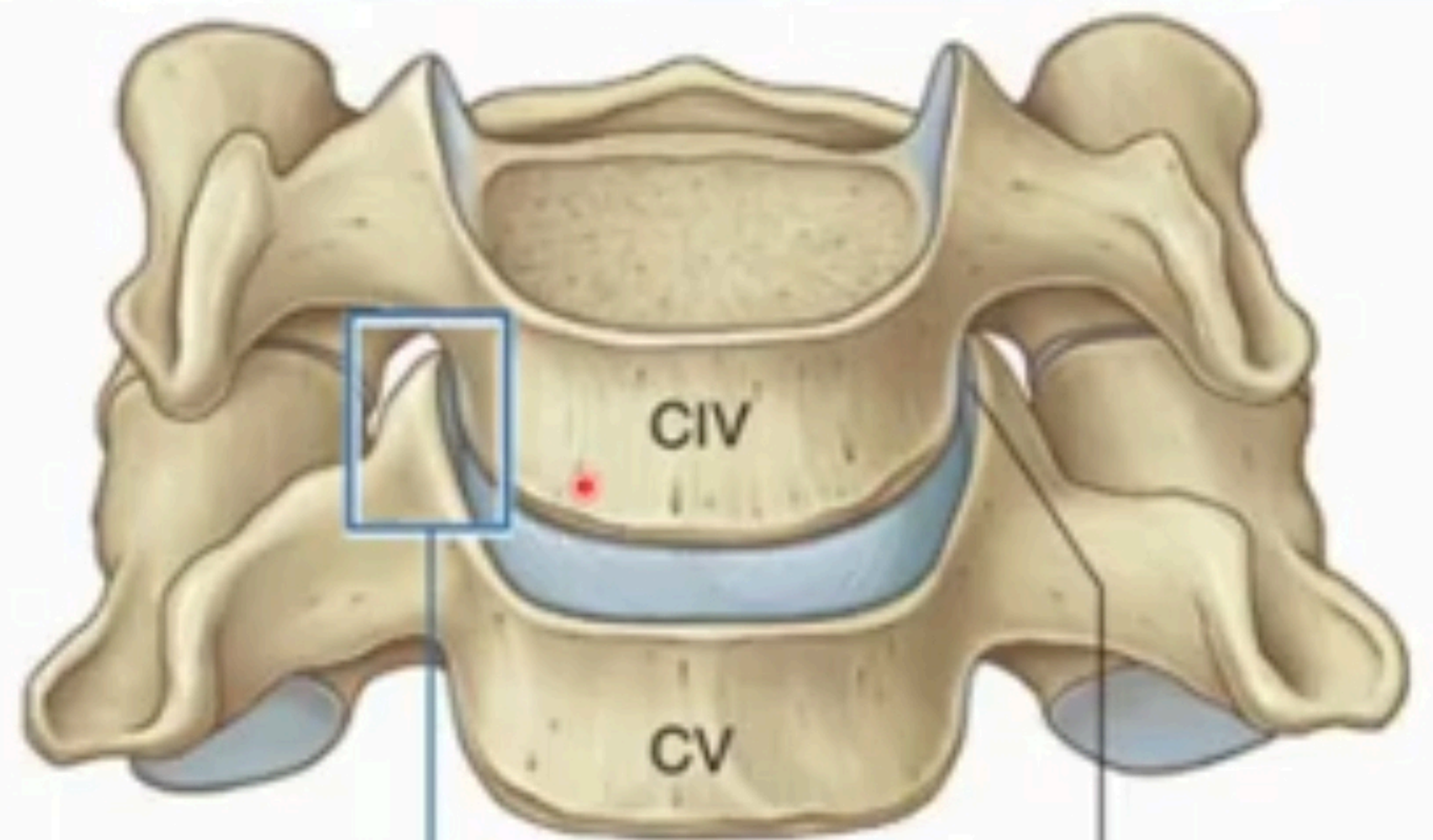


Posterior view

➤ The Joints between vertebrae

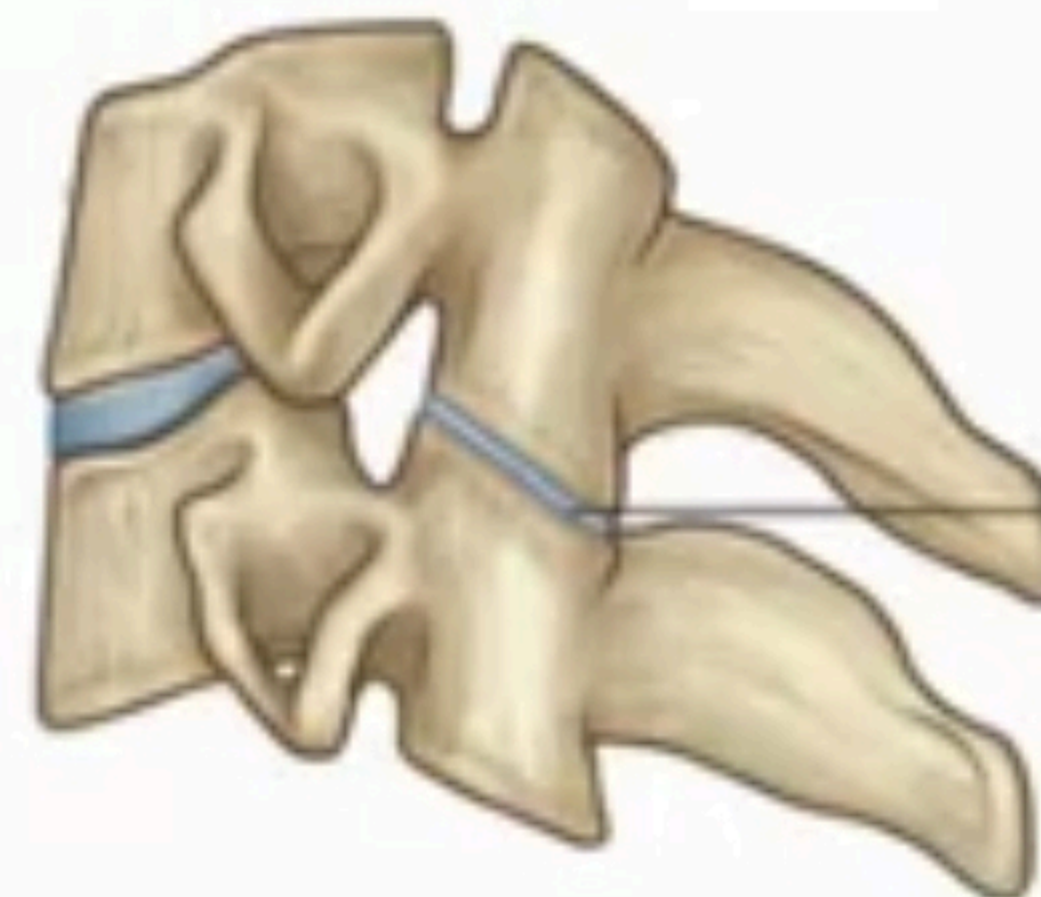


➤ The Joints between vertebrae



Uncovertebral joint

Uncinate process

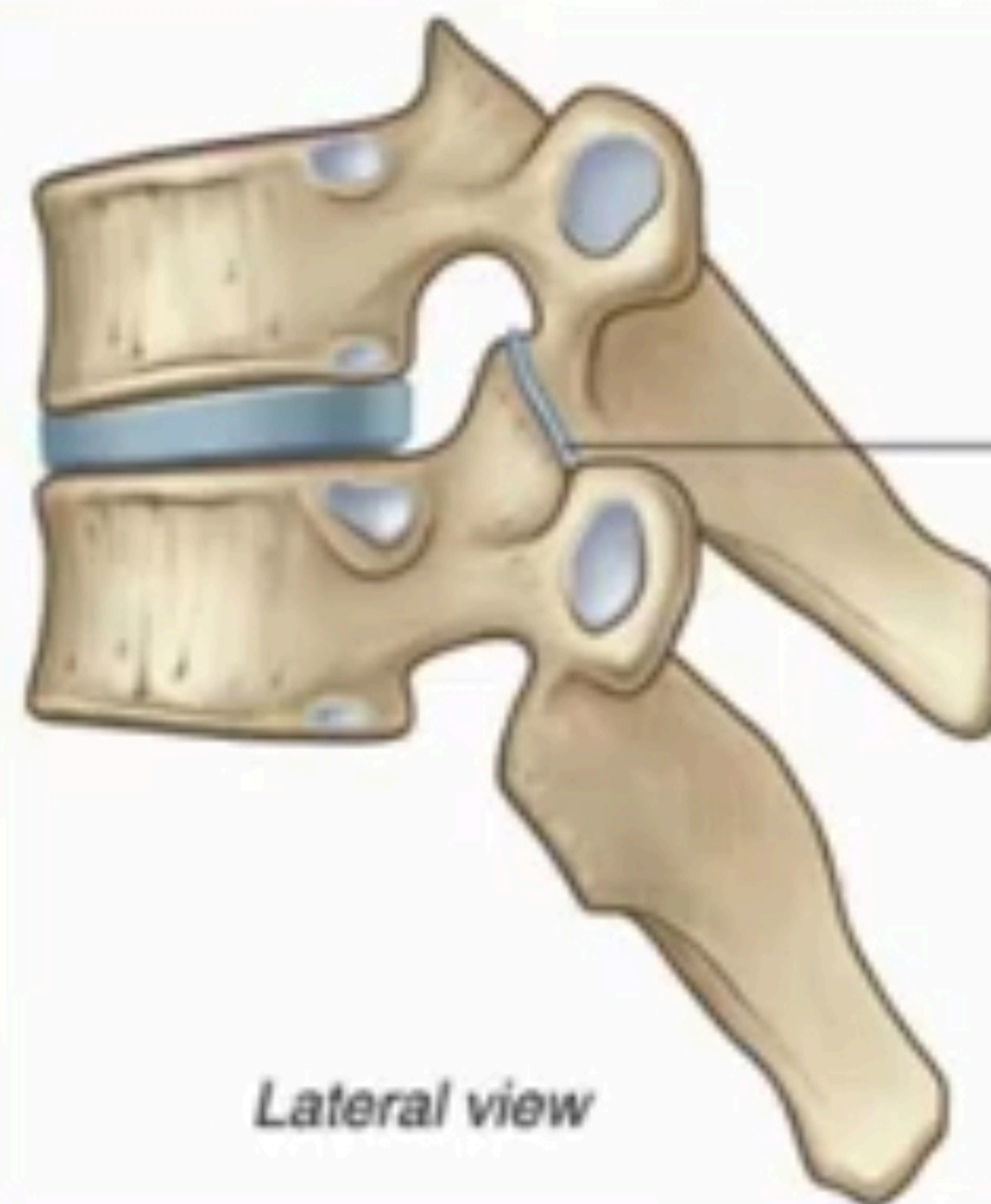


Lateral view

Cervical

"Sloped from anterior to posterior"

Zygapophysial joint



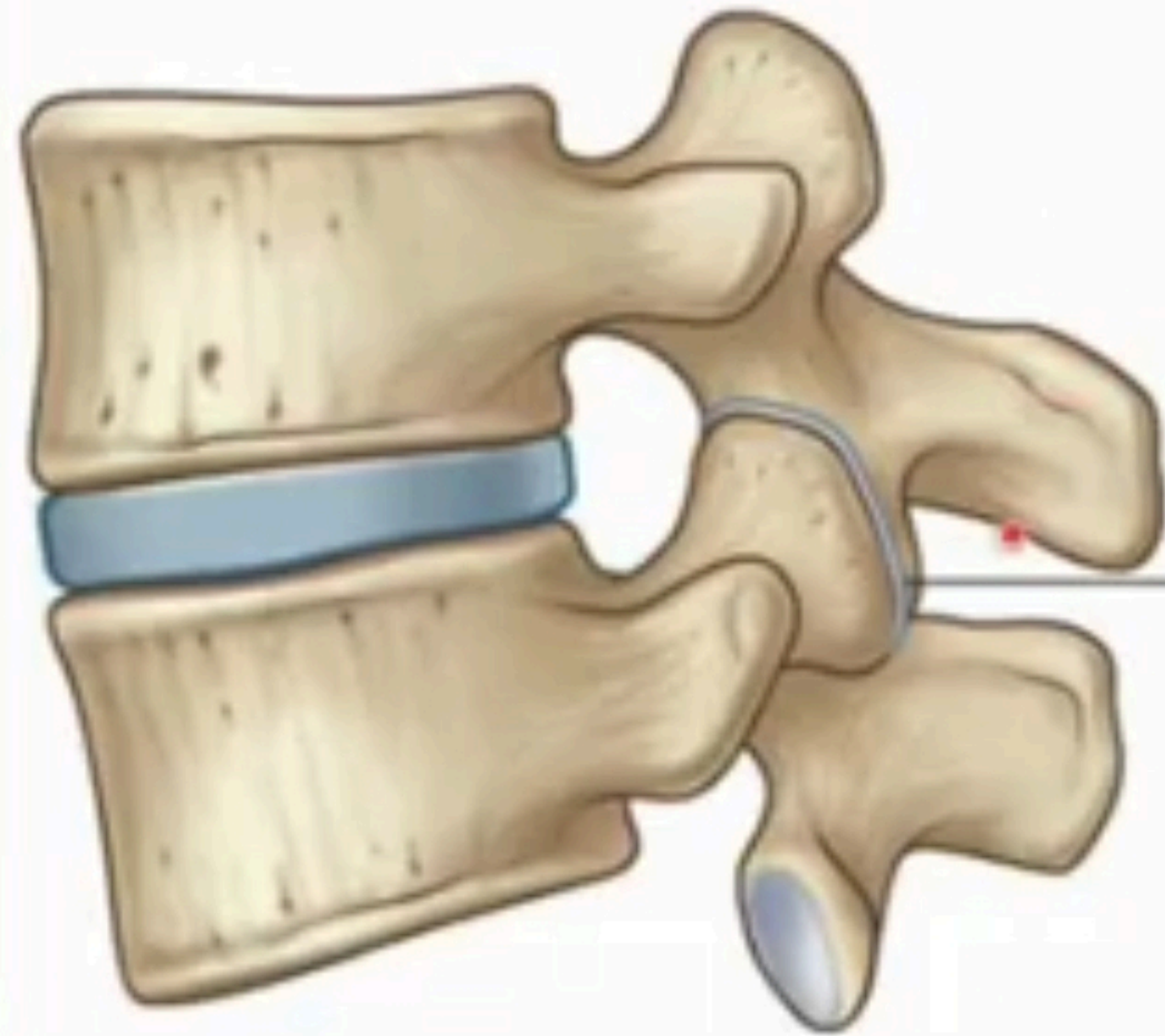
Lateral view

Thoracic

"Vertical"

Zygapophysial joint

➤ The Joints between vertebrae

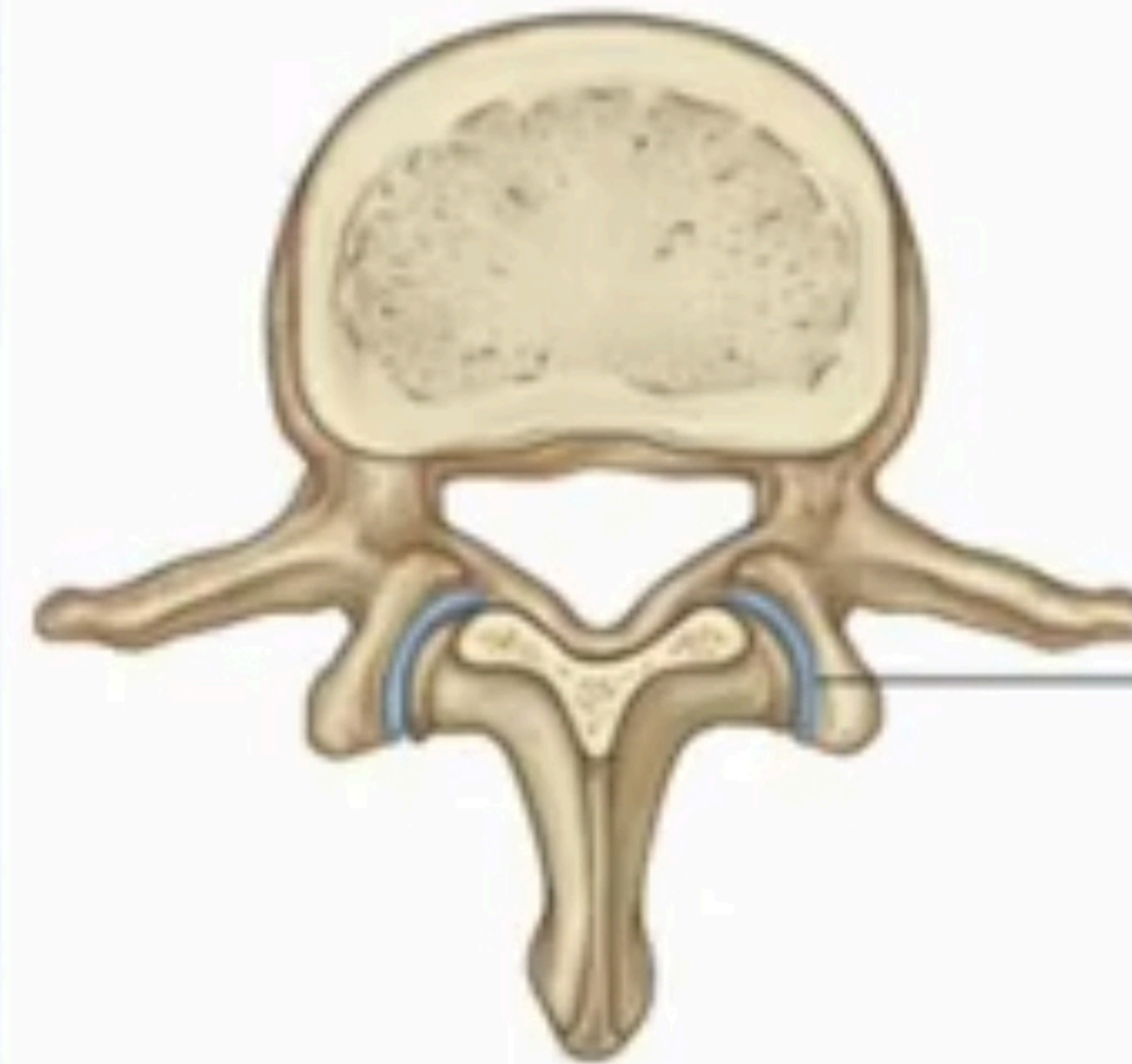


Lumbar
"Wrapped"

Lateral view

Zygapophysial joint

Lateral view



Zygapophysial joint

Superior view

Ligaments between vertebrae

➤ **Supraspinous ligament :**

- ✓ This runs between the tips of adjacent spines.

➤ **Interspinous ligament:**

- ✓ This connects adjacent spines.

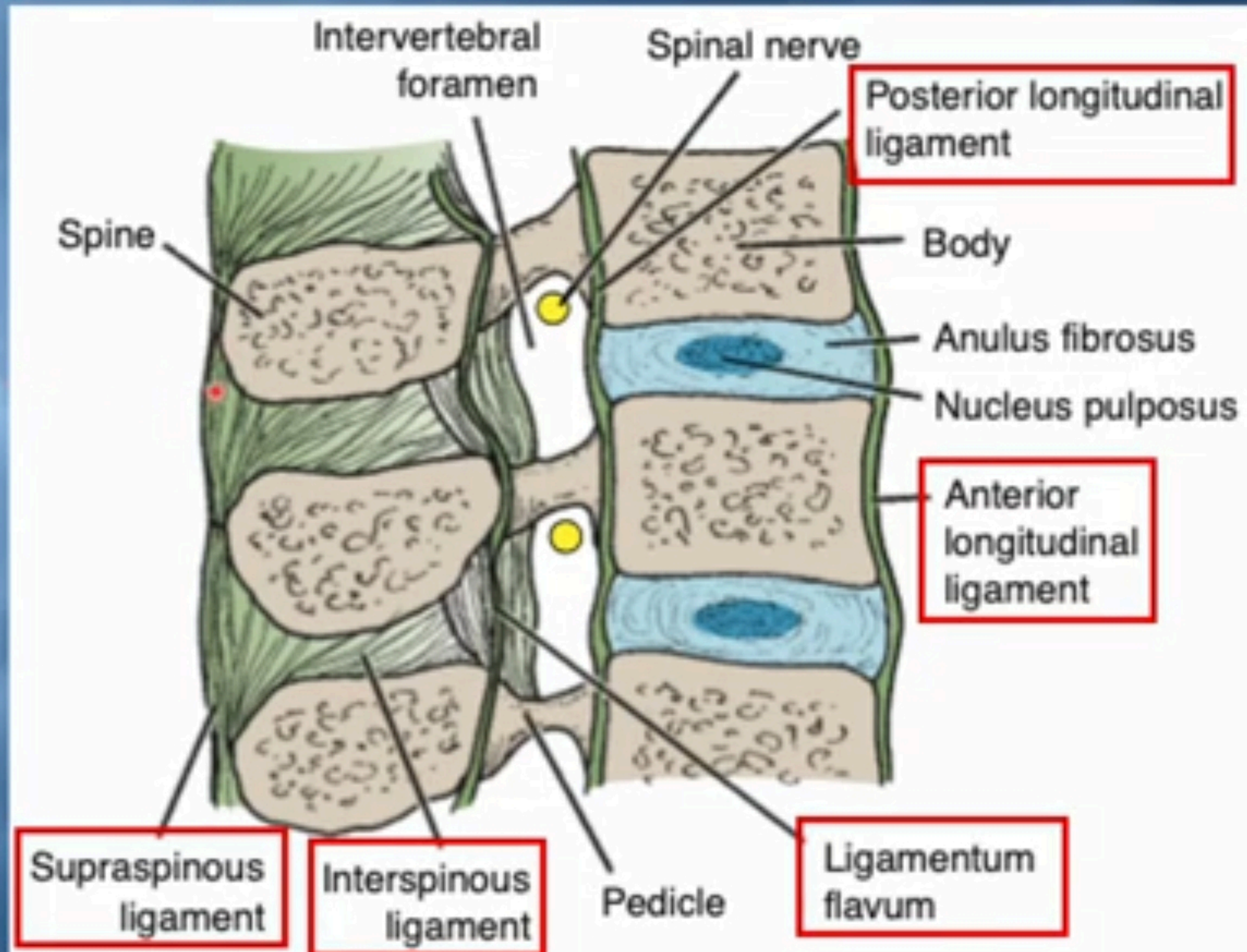
➤ **Intertransverse ligaments:**

- ✓ These run between adjacent transverse processes.

➤ **Ligamentum flavum:**

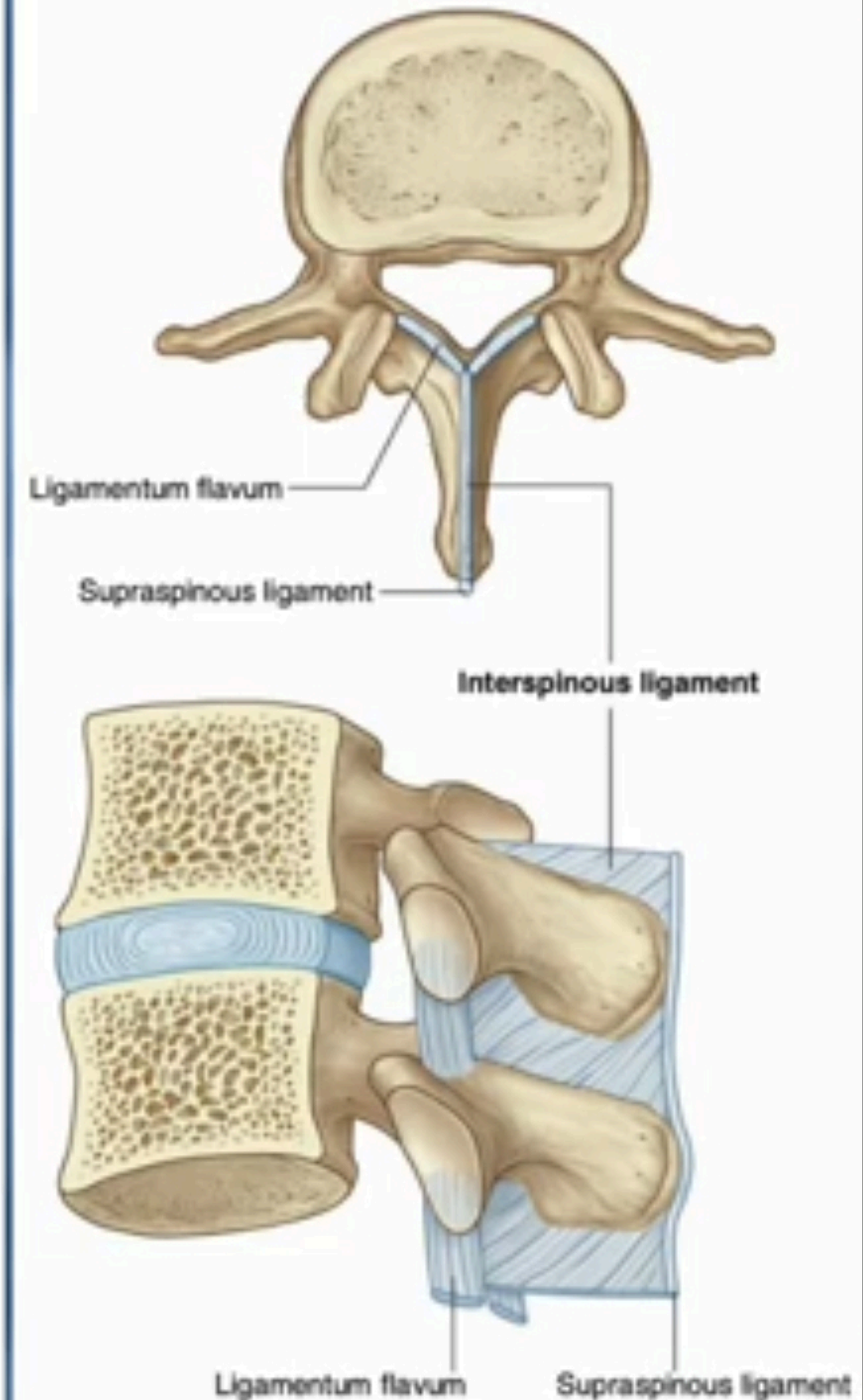
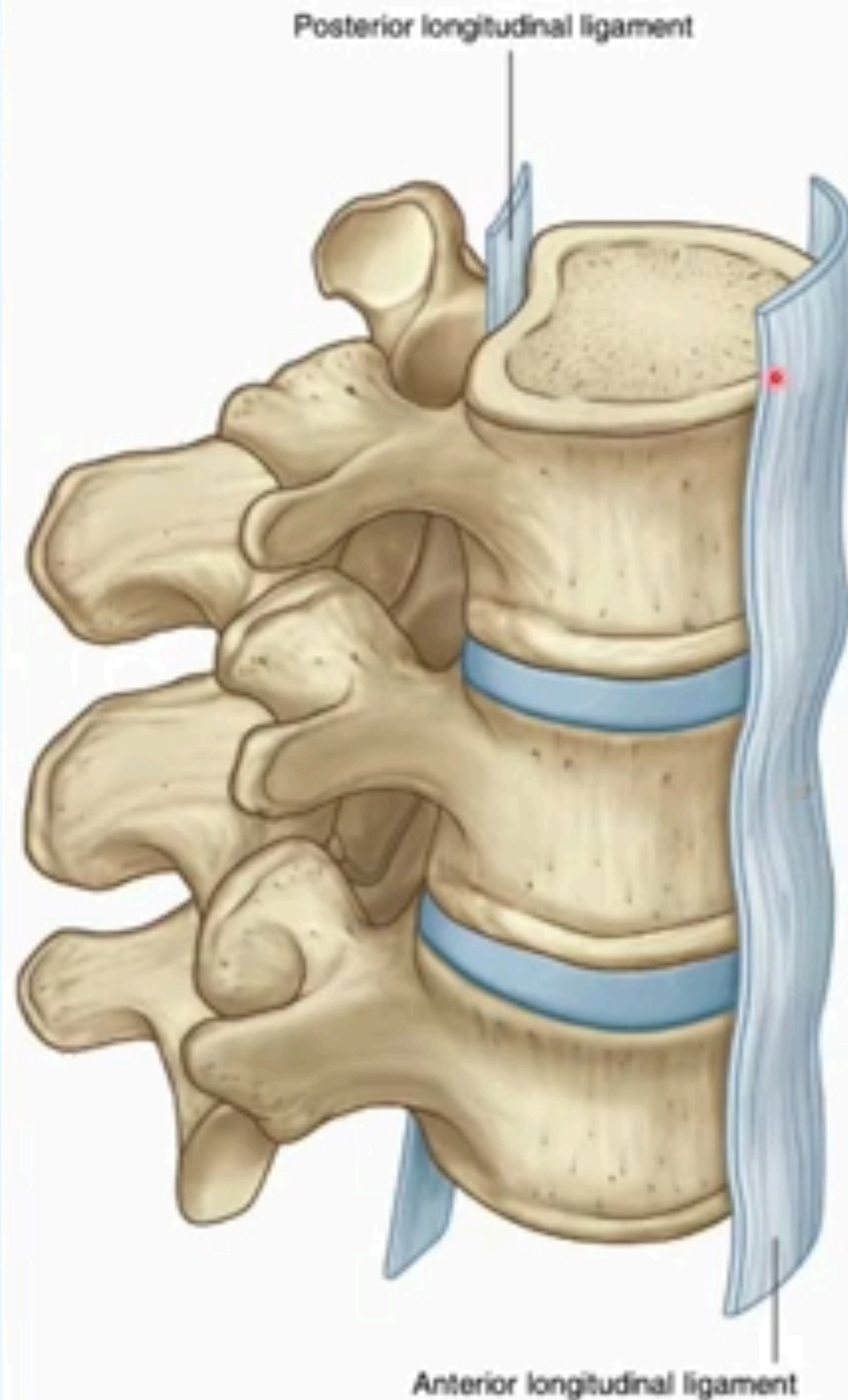
- ✓ This connects the laminae of adjacent vertebrae.

- ❖ In the cervical region, the **supraspinous** and **interspinous ligaments** are greatly thickened to form the **strong ligamentum nuchae**.

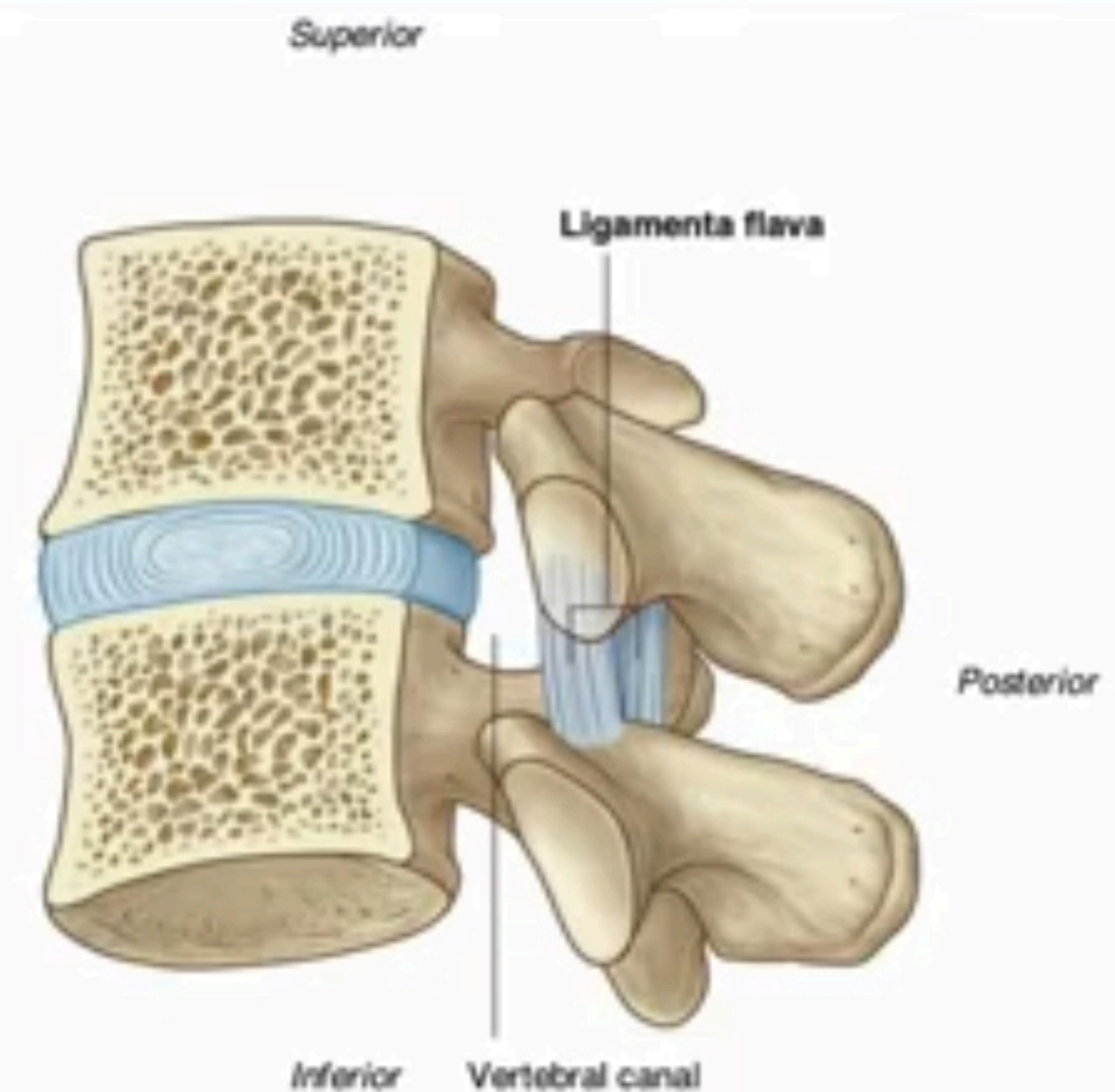
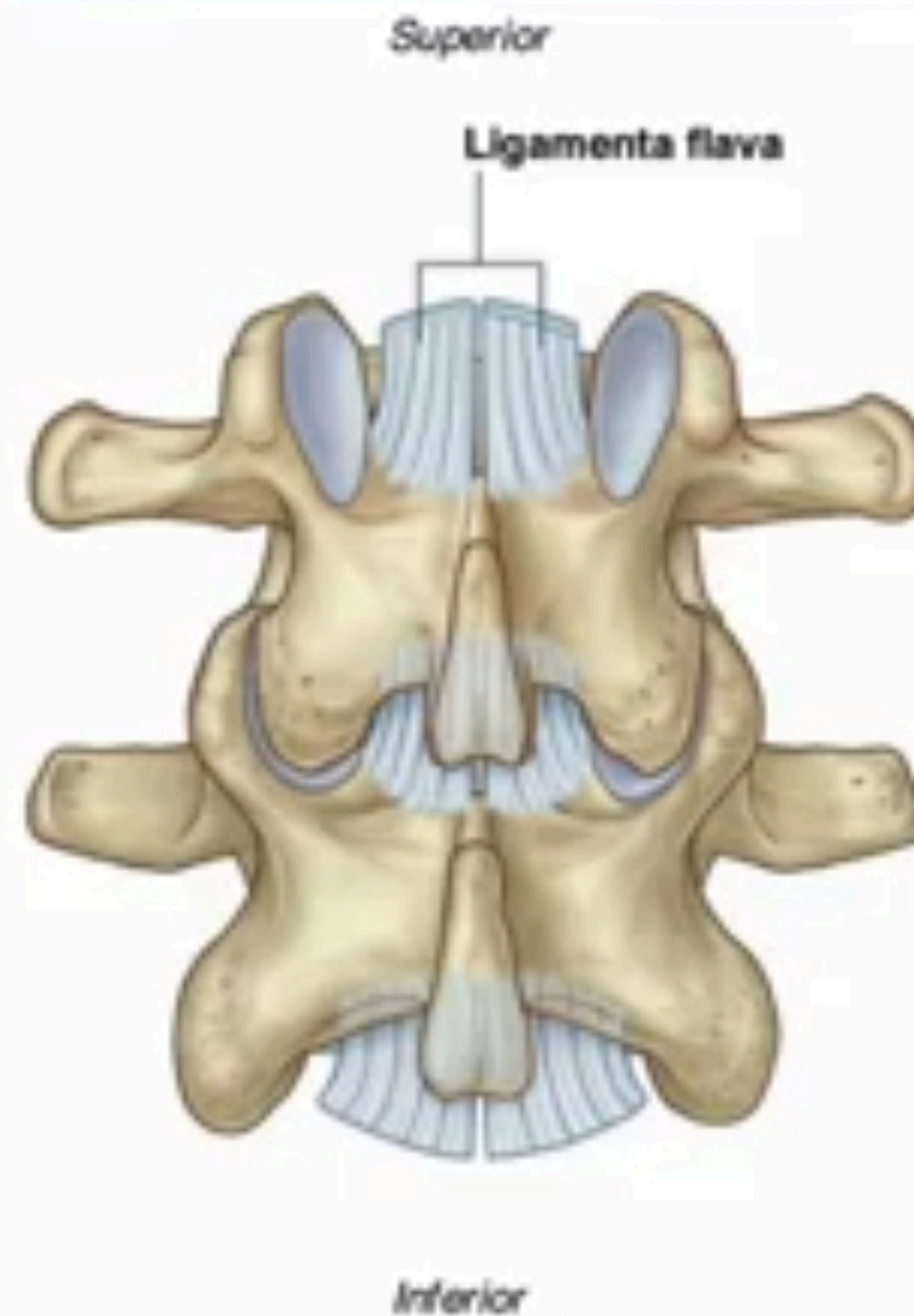
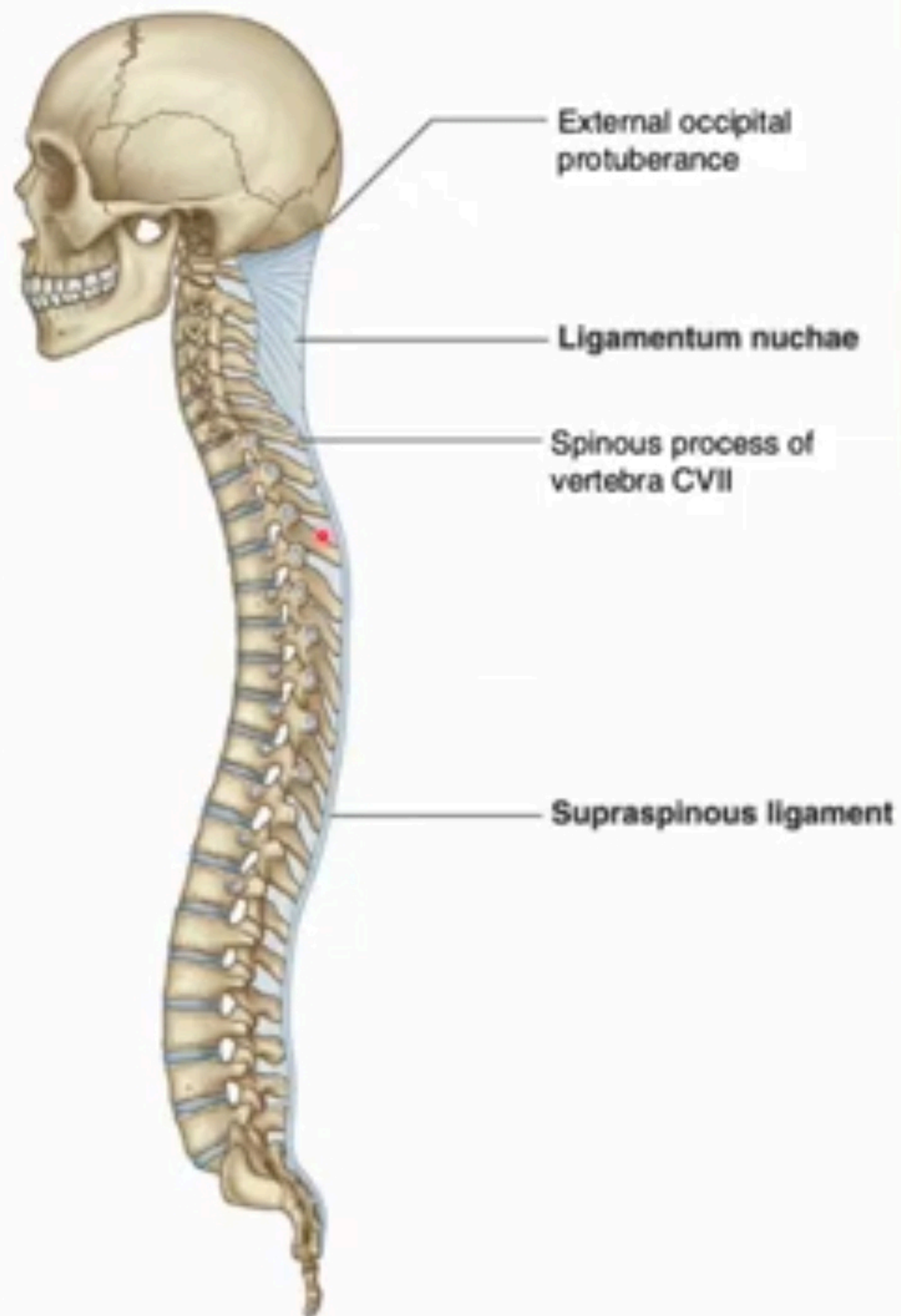


The Ligaments between vertebrae

- Anterior and posterior longitudinal ligment.
- The ligamentum flavum.
- The supraspinous ligament and interspinous ligament.



Ligaments between vertebrae



- Each disc consists of a **peripheral part**, the **anulus fibrosus**, and a **central part**, the **nucleus pulposus**.

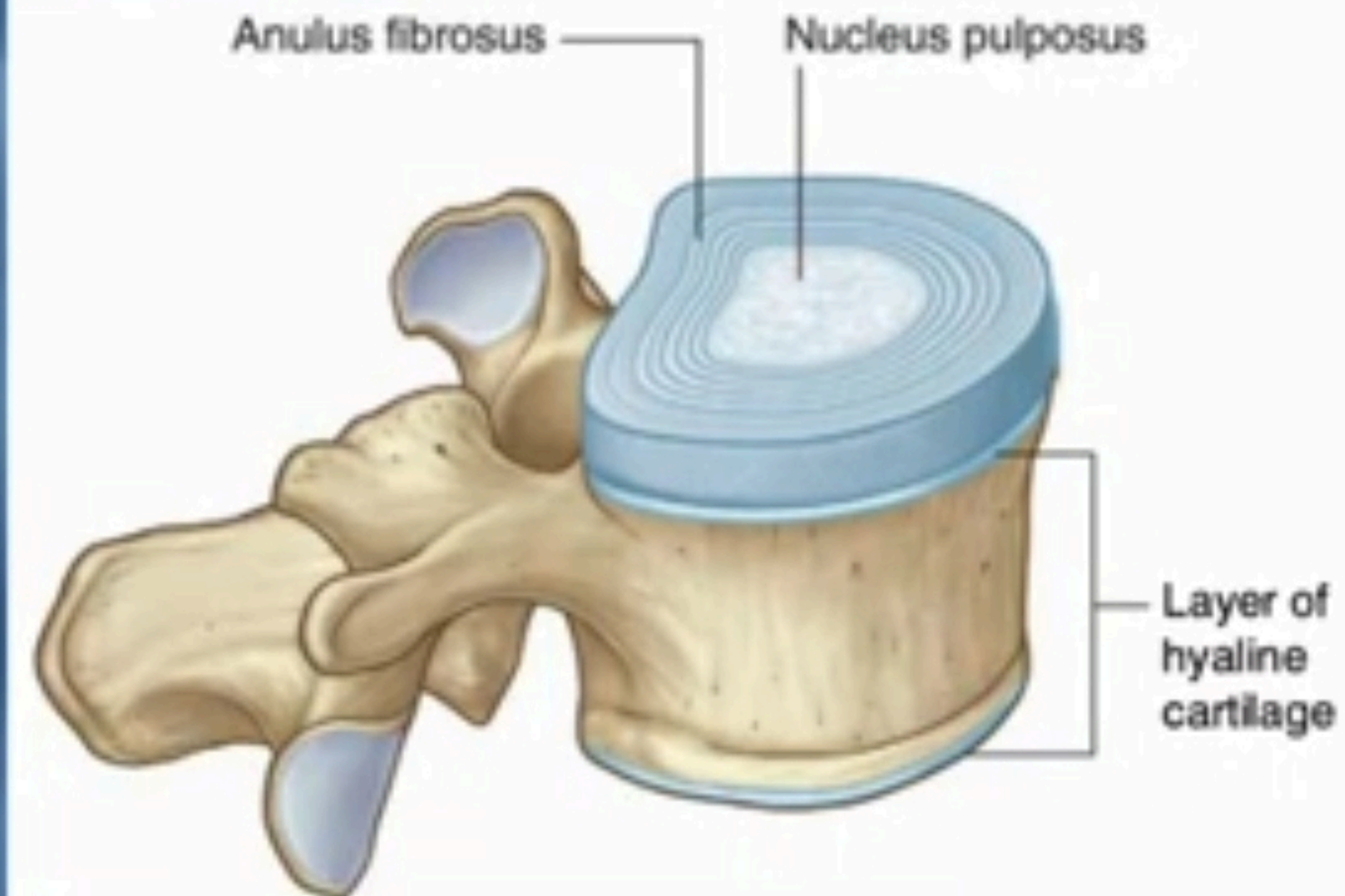
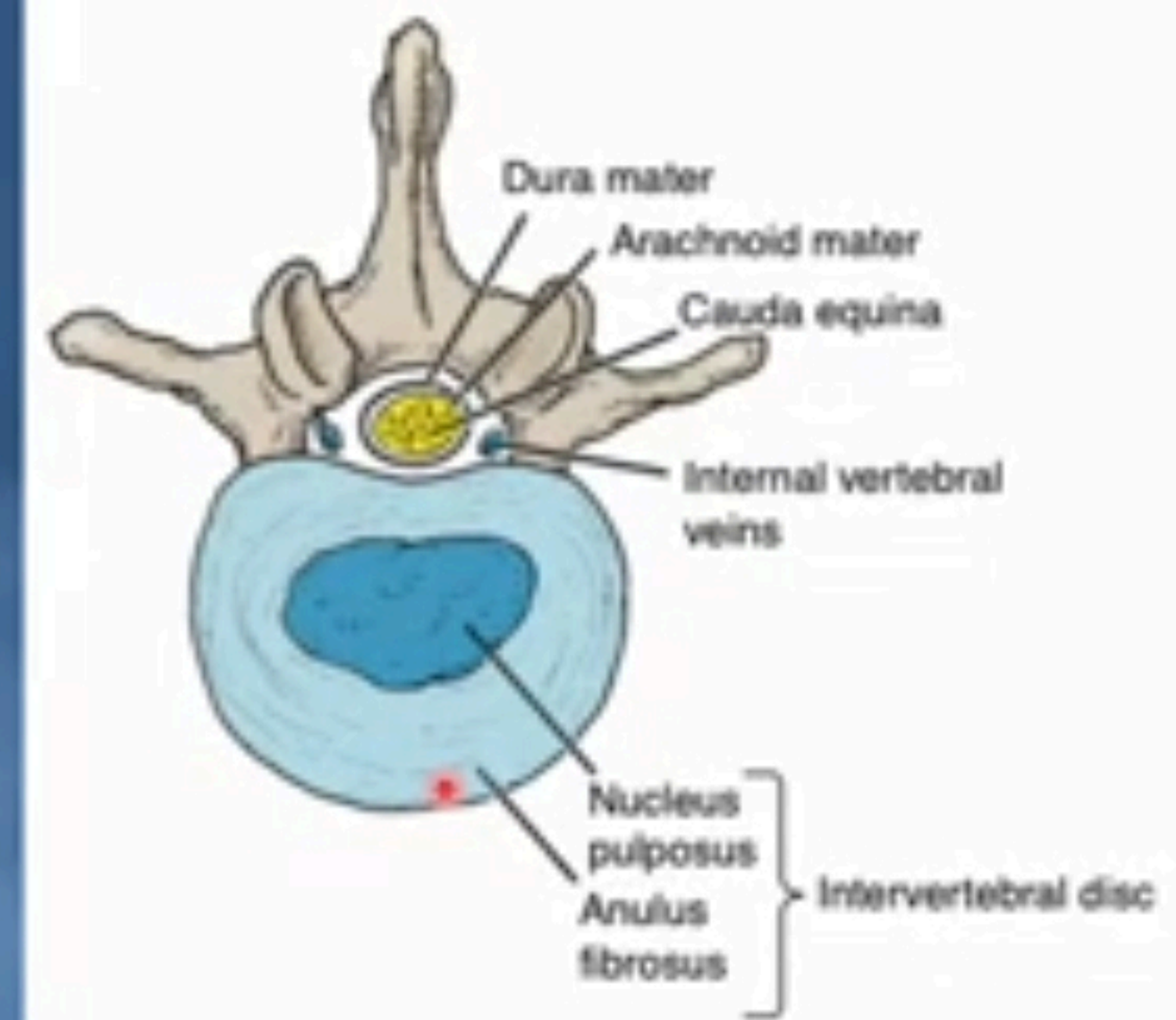
The anulus fibrosus:

- It is composed of **fibrocartilage**.
- It is strongly attached to **the vertebral bodies** and **the anterior** and **posterior longitudinal ligaments** of the vertebral column.

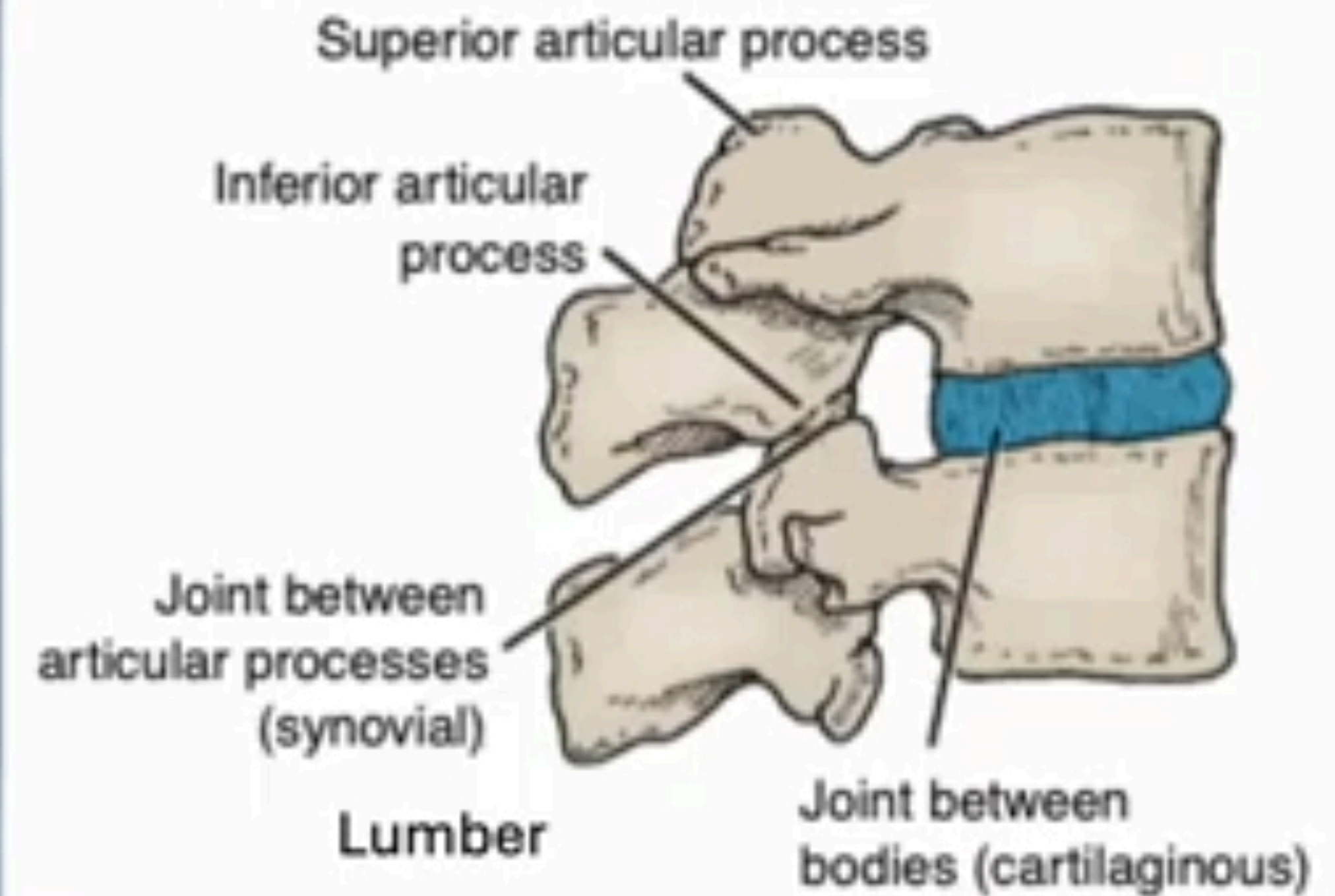
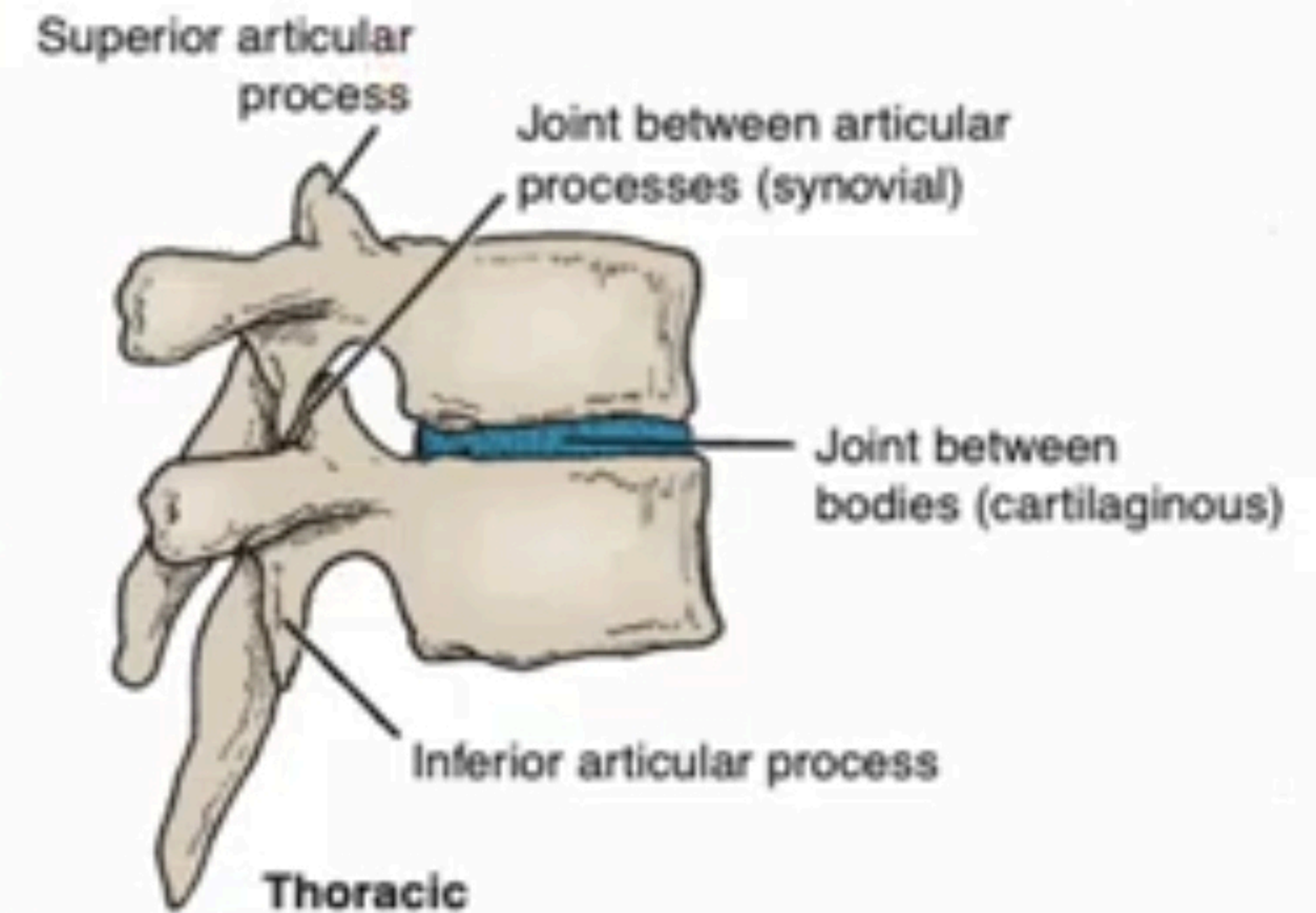
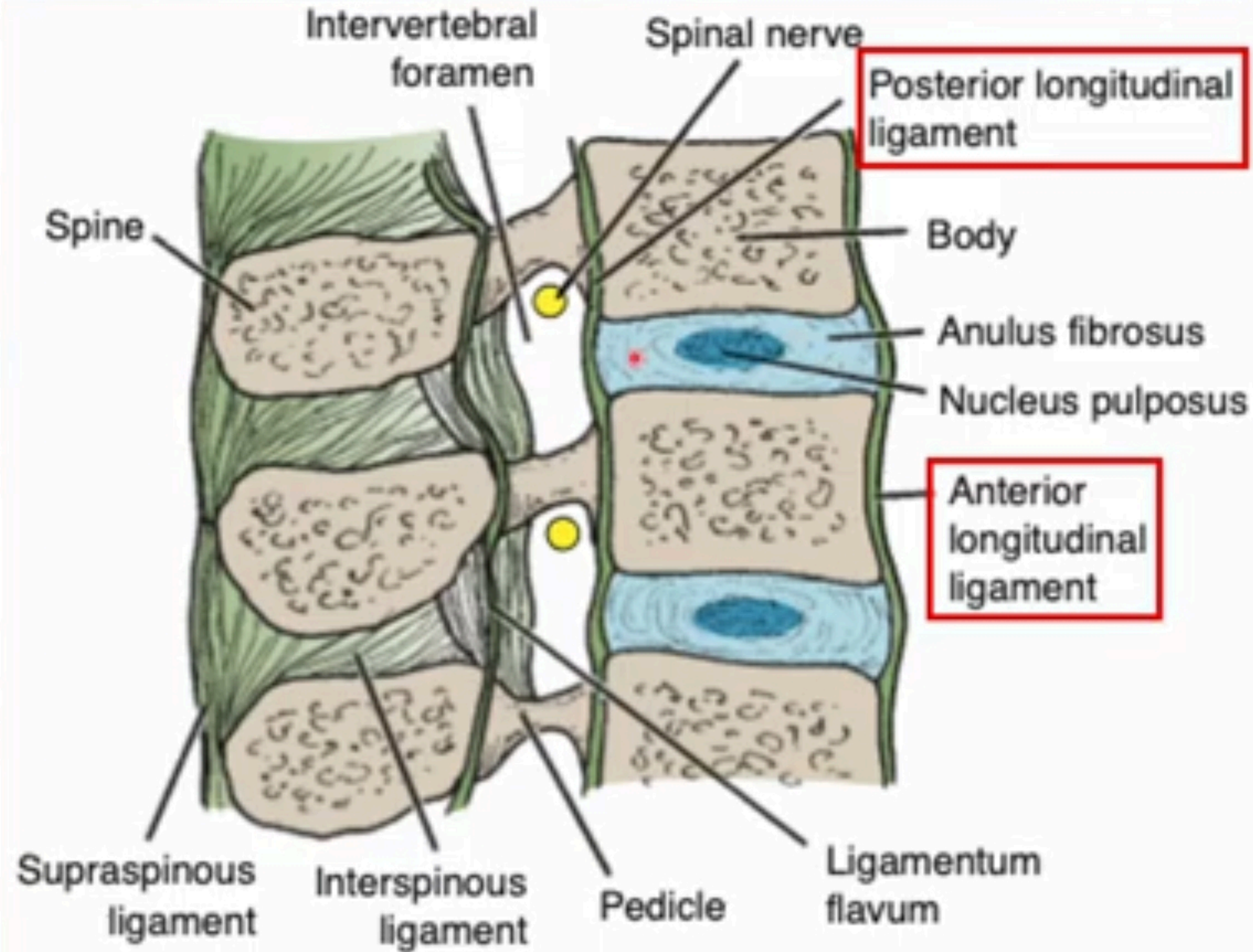
The nucleus pulposus:

- It is an **ovoid mass** of gelatinous material.
- It is **normally under pressure**.
- The upper and lower surfaces of the bodies of adjacent vertebrae that abut onto the disc are covered with thin plates of **hyaline cartilage**.

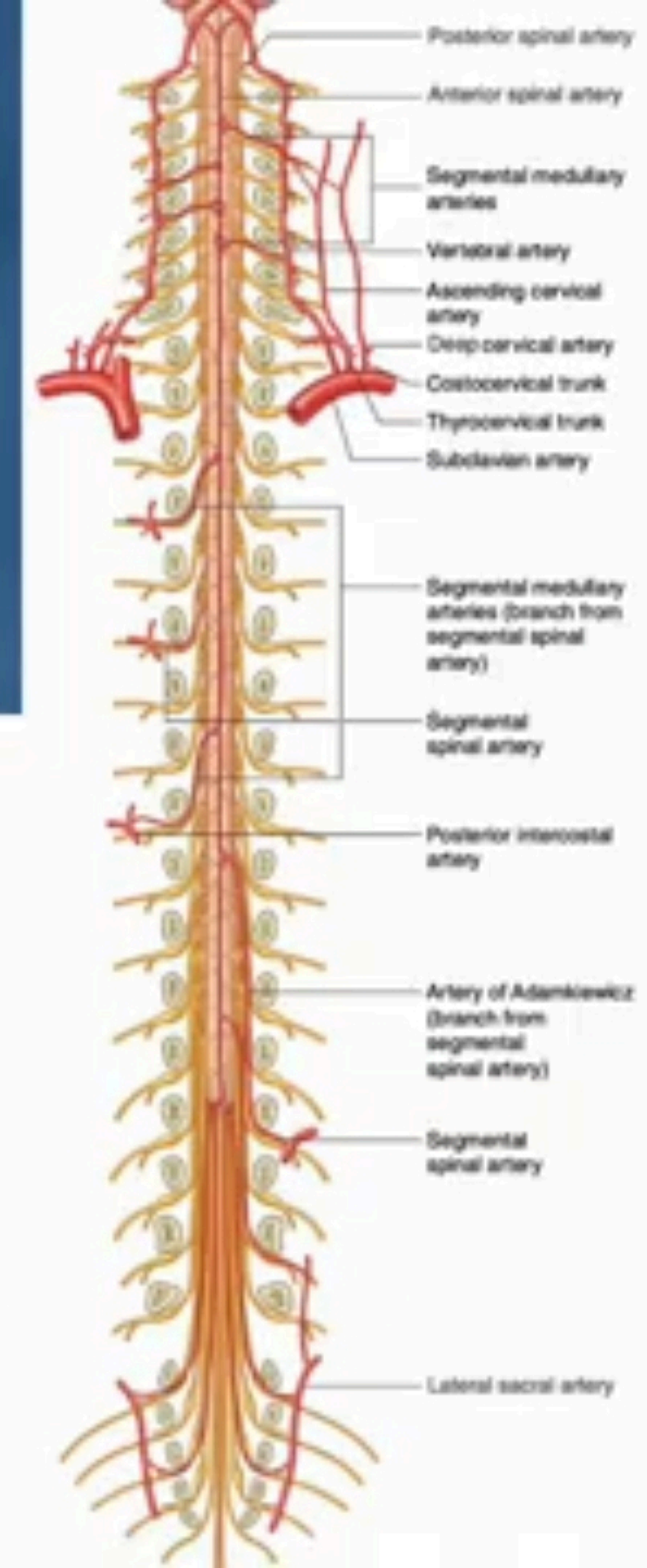
The disc



The disc



Blood supply of vertebra and spinal cord



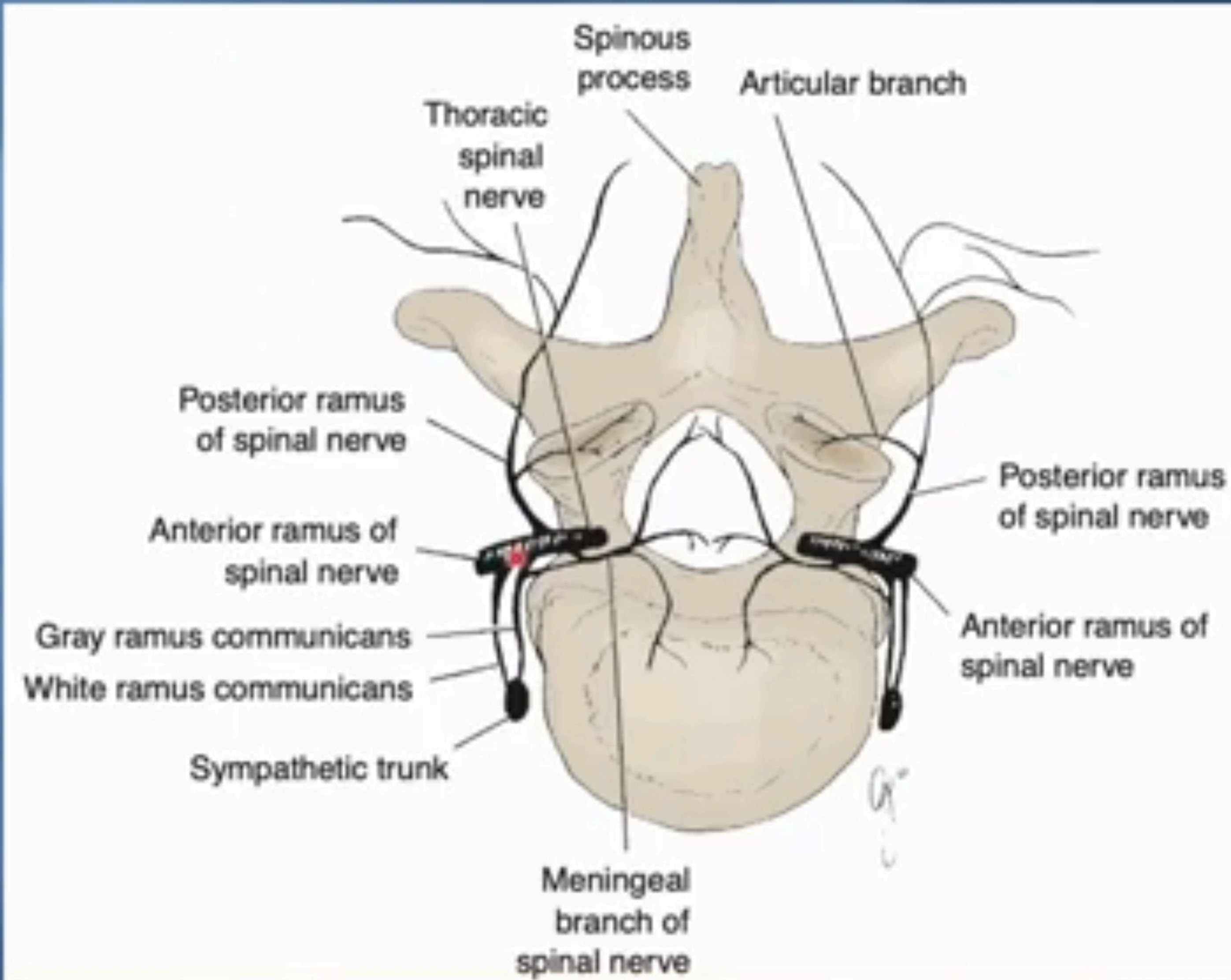
Vertebral arteries travel in transverse processes of C6-C1, then pass through foramen magnum

Cervical region

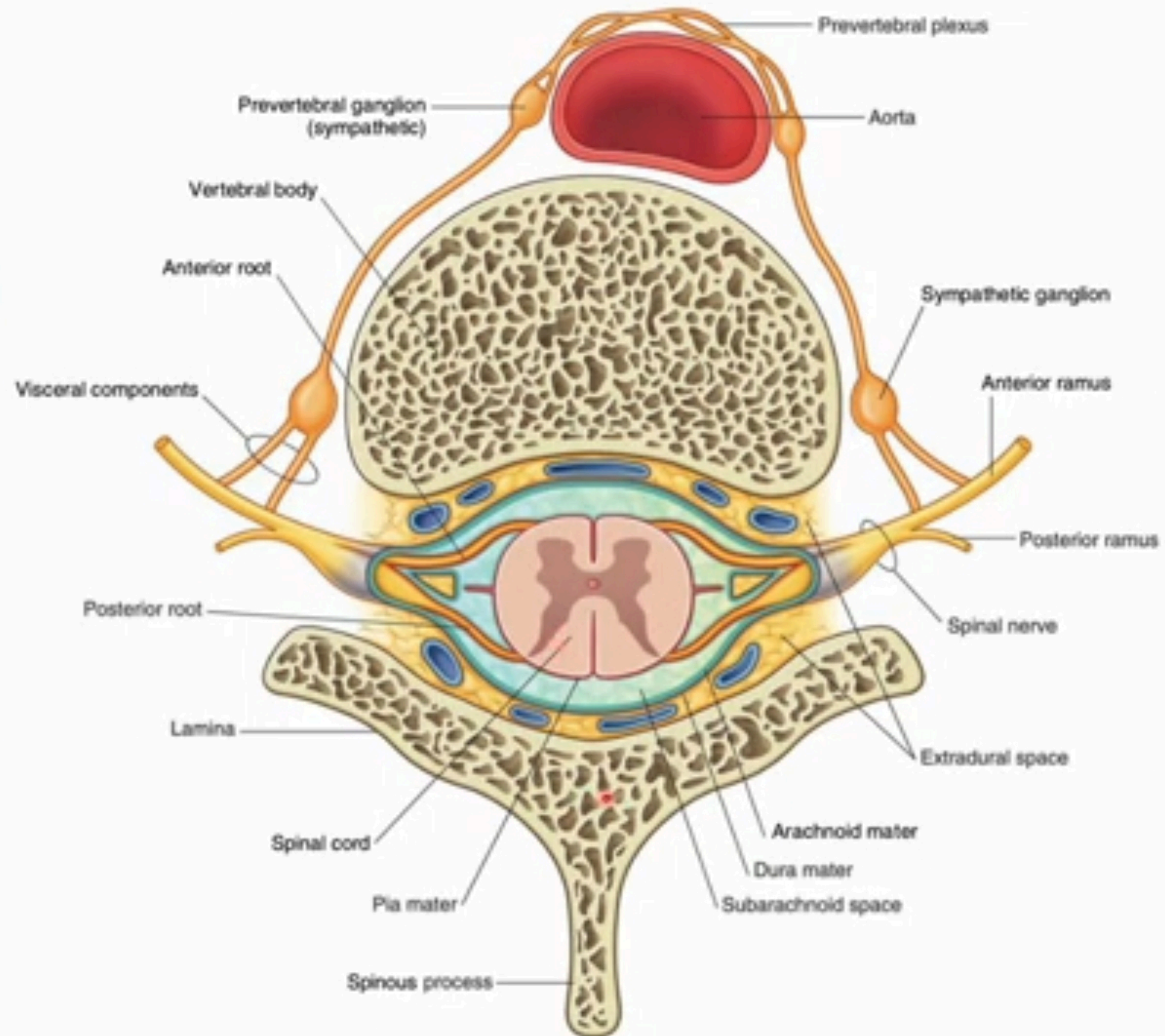
- supports and moves head
- transmits spinal cord and vertebral arteries between head and neck

Nerve Supply of Vertebral Joints

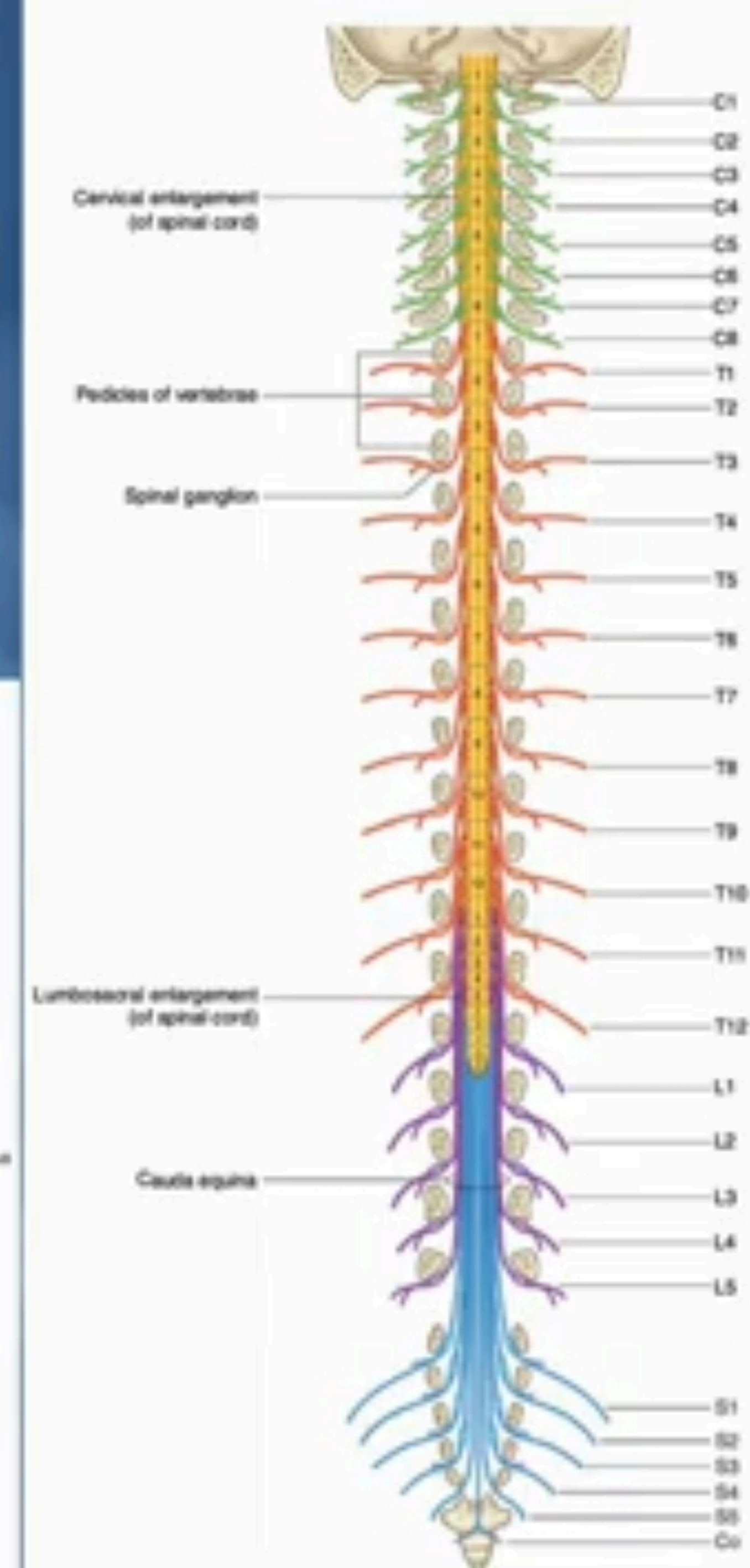
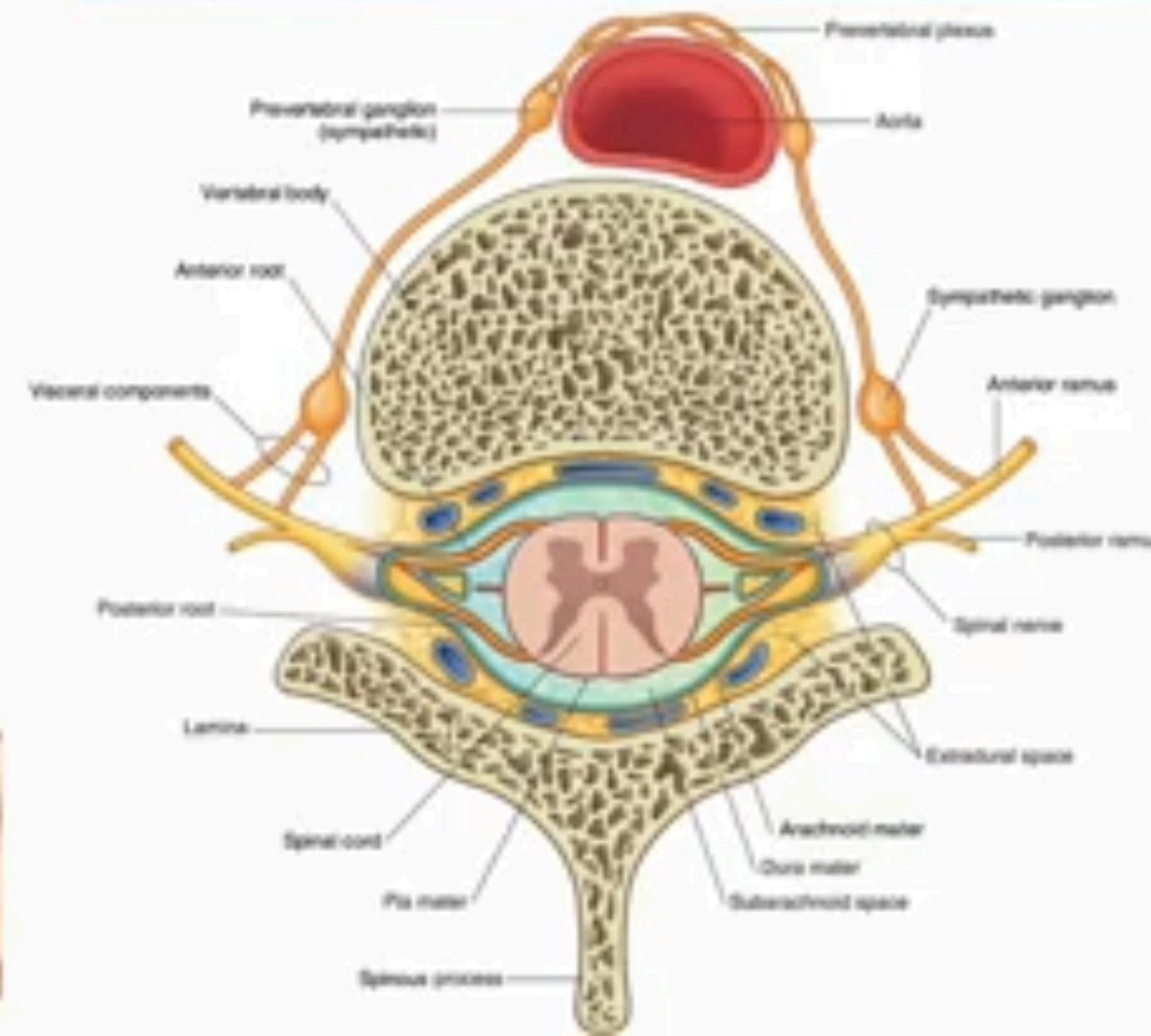
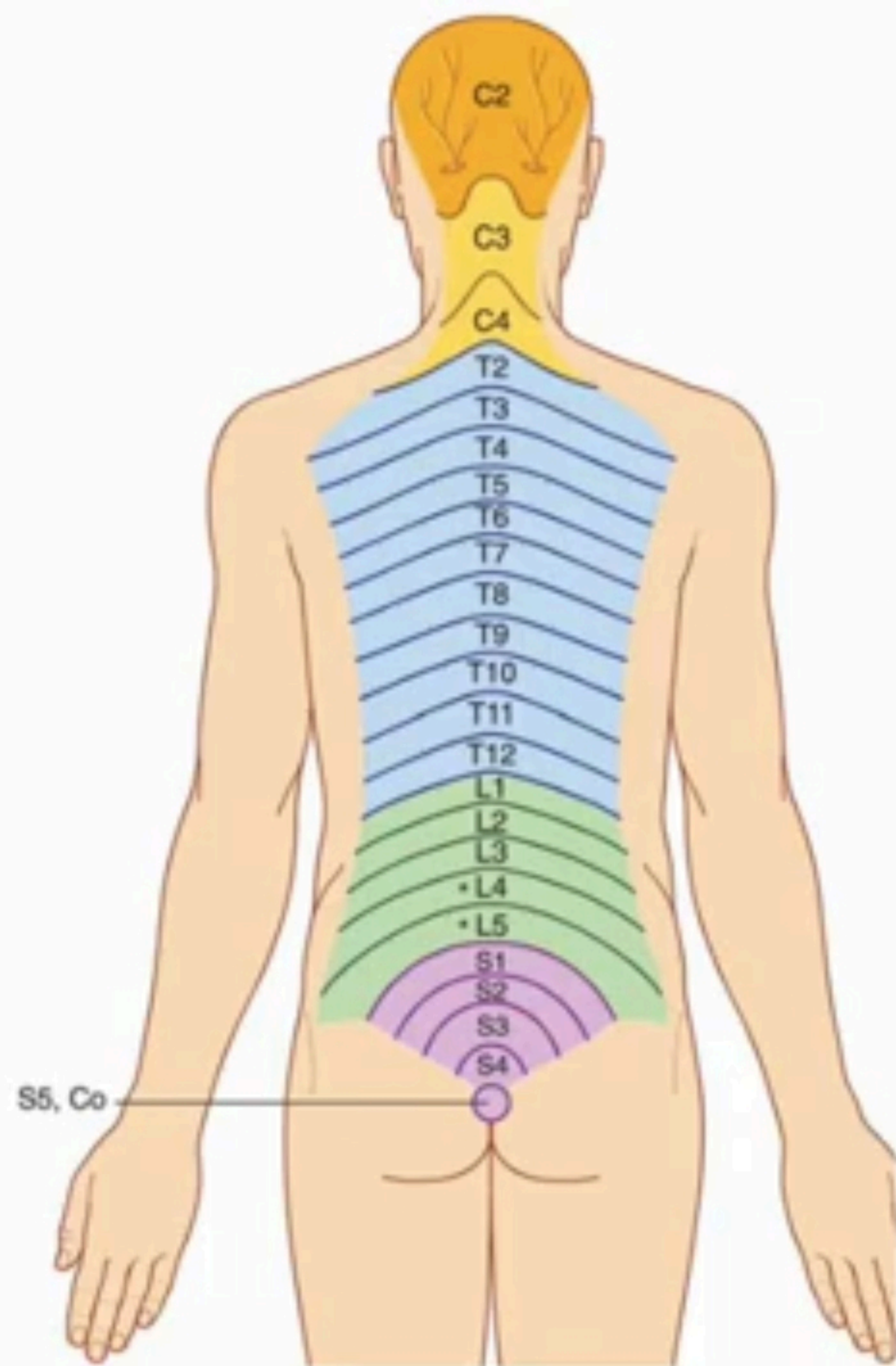
- The joints between the vertebral bodies are innervated by **the small meningeal branches** of each spinal nerve.
- The joints between the articular processes are innervated by **branches from the posterior rami of the spinal nerves** the joints of any particular level receive nerve fibers from two adjacent spinal nerves.
- The atlanto-occipital joints and the atlanto-axial joints should be reviewed in a textbook of gross anatomy.



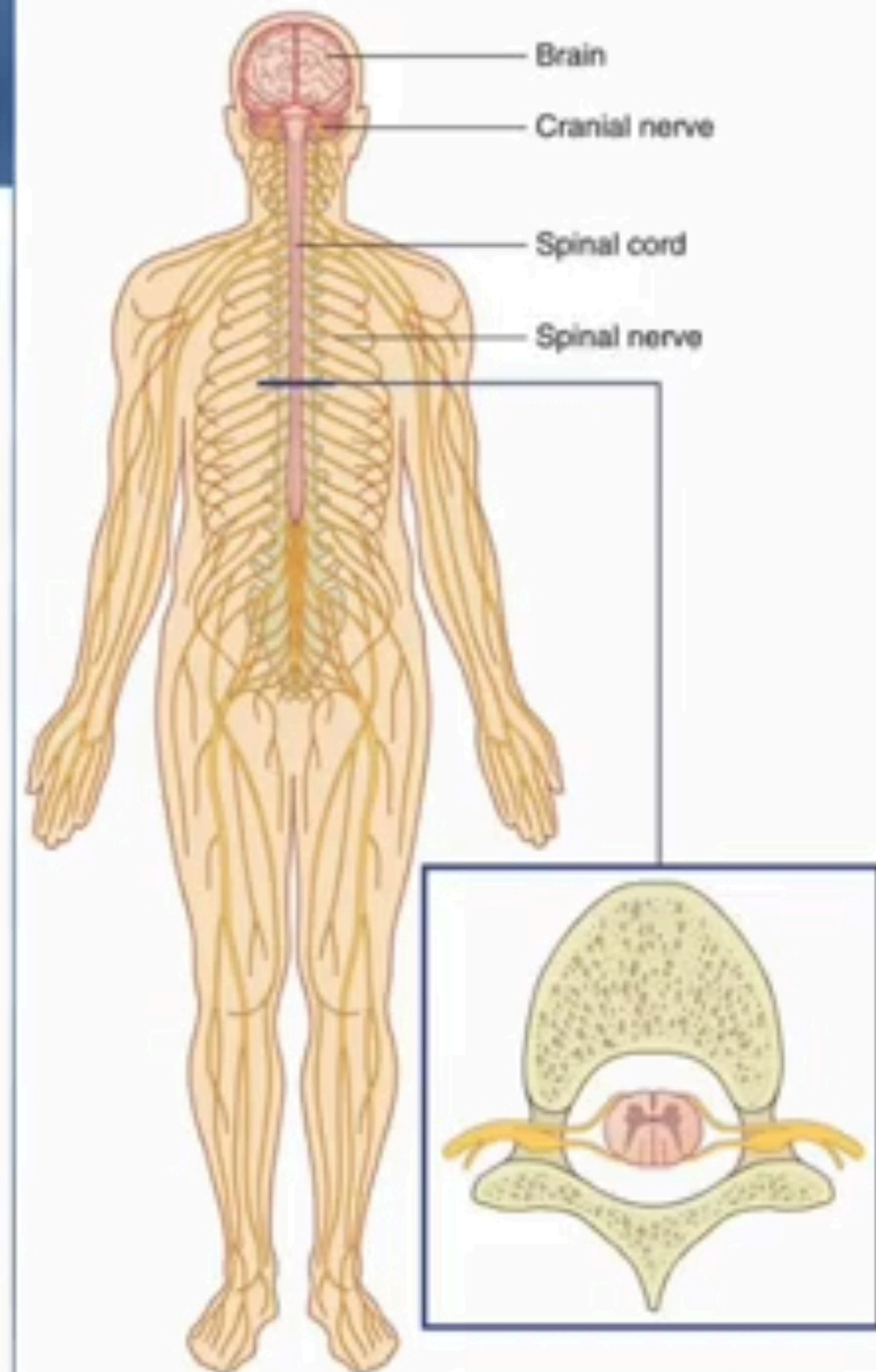
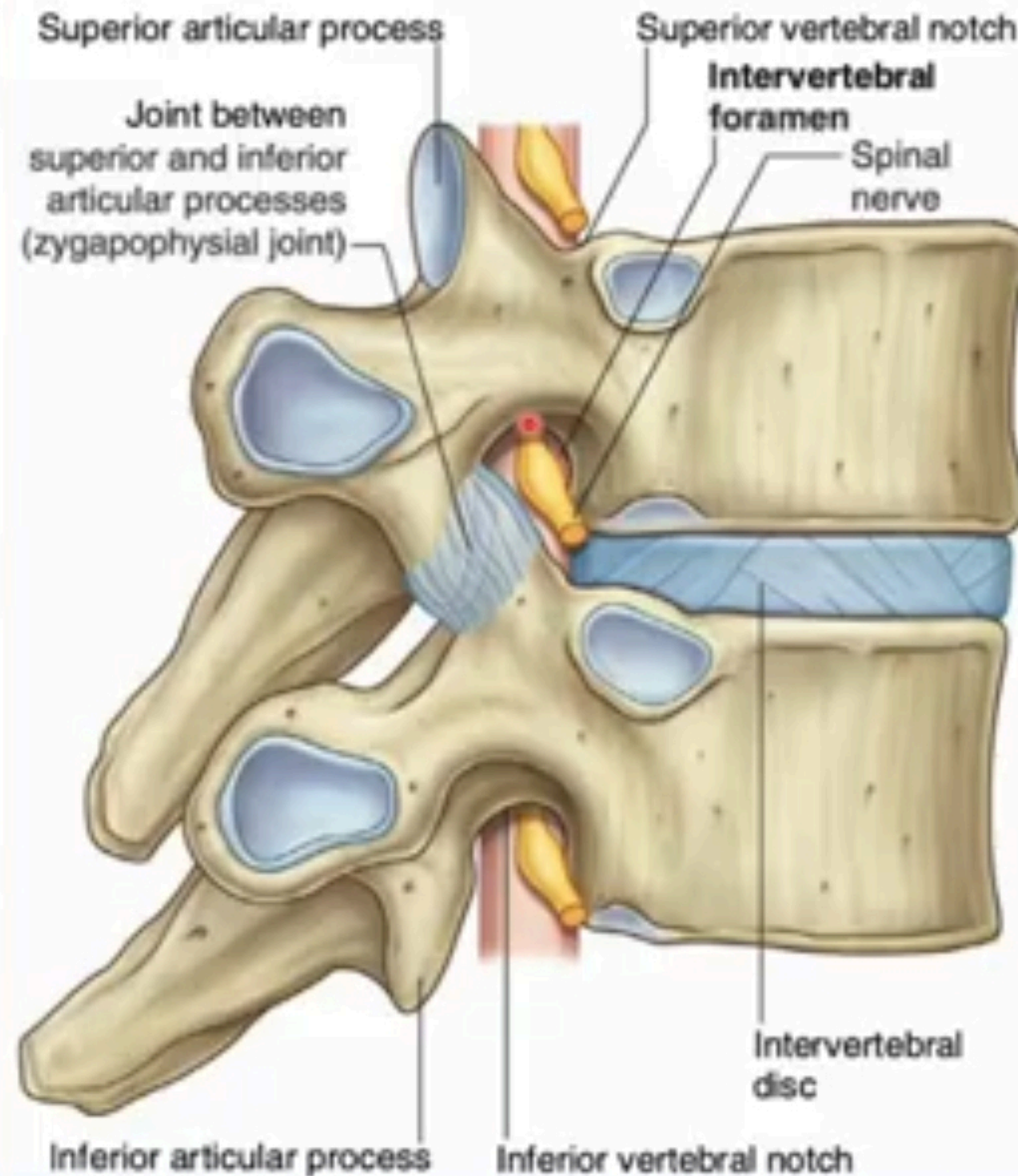
Nerve Supply of Vertebral Joints



Dermatomes innnervated by posterior rami of spinal nerves



Intervertebral foramen



Will be continue ...

